

NEURAL BELLMAN OPERATORS

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We develop Neural Bellman Operators for continuous-time economic models with recursive utility, costly preference adjustment, and strategic interaction. A neural critic evaluates a policy, a feasible actor improves its Hamiltonian, and independent verification bounds economic error. We provide differential and monotone Bellman formulations, guarded evaluation, and complete-action error bounds. A performance-difference theorem links costate regression, rollout bias, and actor optimization to welfare. In a dense nonlinear capital economy, policy-specific certificates combine common-path payoff comparisons with diffusion-transfer and sampling bounds on an unbounded state space. Implementation results cover exact state histories and finite, noisy state observations. Executed experiments compare NBO with direct policy optimization, affine rules, raw-costate improvement, and independent scalar references, reporting gains, accuracy, and computational work. Preference and game applications show how the same evaluation and improvement structure produces distinct welfare and unilateral-deviation accounts.

KEYWORDS: Dynamic programming, Neural approximation, Recursive utility, Endogenous preferences, Viscosity solutions, Dynamic games.

1. INTRODUCTION

Quantitative economic models increasingly place several margins of adjustment inside the same dynamic decision problem. A household chooses consumption and portfolio risk while investing in its own preferences. Recursive utility makes continuation utility part of the policy-evaluation equation. A sophisticated consumer anticipates the decisions of future selves, while a firm invests in anticipation of its rivals' policies. These problems require a computational method that evaluates the relevant continuation values and determines whether feasible changes in behav-

ior improve the stated economic objective. The motivating preference-formation problem of Qi (2025) brings the internal and external margins together.

This paper develops Neural Bellman Operators (NBO) around policy evaluation, feasible improvement, and economic verification. A neural critic represents continuation utility, and a feasible actor represents the controls entering the state dynamics. The critic evaluates a fixed policy using its boundary conditions and utility domain. The actor then improves its Hamiltonian with the critic and its derivative channels held fixed. Independent verification measures the remaining evaluation error and the gain from feasible deviations. These steps give the trained networks an economic interpretation and specify the accuracy of the policy actually implemented.

Our first contribution is a common evaluation and improvement formulation for the economic applications. In smooth problems, a verification theorem translates boundary error, fixed-policy residual error, and feasible-action error into value and policy-loss bounds. For nonsmooth problems, a positive-weight Bellman formulation supplies the monotonicity needed for viscosity selection. A guarded evaluation map enforces a declared nodal tolerance and records the corrections required to do so. Complete-action bounds use signed derivative enclosures and verified face domination, including the effects of stopping switches. The representation can therefore be chosen for numerical flexibility while the economic acceptance criterion is expressed in the model's values and deviations.

The second contribution carries policy-specific verification through to a dense nonlinear capital diffusion on its original unbounded state space. A feasible analytical schedule supplies an absolute comparison with the economic optimum. Curvature controls the benefit from changing consumption, and a spectral bound controls the propagation of those changes through the coupled production process. A paired payoff identity then measures the value of a fitted correction to the schedule. Explicit transfer, tail, arithmetic, and sampling terms turn this comparison into a policy-specific continuous-time bound. The comparison class includes the original adapted controls. The candidate controller is specified by its update rule, memory, and information structure, so that the evaluated policy and the implemented policy are the same mathematical object. A product-space argument shows that

independent private randomization preserves the economic value. An observation construction implements the internal controller from exact state histories, and a separate payoff-transfer theorem extends implementation to finite dates and noisy state measurements.

The third contribution makes the role of the Bellman evaluation block testable. In the capital economy, strong concavity bounds Hamiltonian loss by costate error and actor suboptimality. A conditional-projection identity separates variation in rollout costates from the error of a learned predictor. A performance-difference theorem integrates the resulting decision losses along the candidate policy's state distribution and separates costate regression error, rollout bias, and actor suboptimality in a welfare-improvement inequality. These results identify the quantities through which critic learning can improve an actor. The numerical study accordingly compares NBO with direct policy optimization, affine policies, and critic ablations. Direct method differences are formed on common simulated paths, with each fitted candidate fixed before final testing. Comparisons report economic endpoints, candidate-generation work, and verification work; checkpoint frontiers expose the relation between accuracy and computation. The recorded initialization streams describe the fitted policies in the experiment, and the stated confidence event concerns their independently evaluated payoffs.

The original applications give these computational objects distinct economic meanings. In the preference model, value gradients determine consumption and the costly adjustment of risk aversion, while a cross derivative links portfolio demand to preference risk. The policywise cost comparison establishes value monotonicity in the adjustment cost, and a specified compensating transfer gives a welfare interpretation to numerical error. Recursive utility is evaluated within a declared invariant utility domain. For temporal selves, the actor satisfies a spike-deviation condition against two endogenous continuation values. In the capital game, the independent criterion is a complete unilateral best response against fixed rival policies. The main text states this application structure, and the supplement gives the full models, derivations, and numerical records. Each conclusion is attached to the economic domain and approximation for which it is established.

The capital study also supplies two economically interpretable comparisons. A finite population of initial balance sheets varies aggregate capital and cross-sectional dispersion, while separate state stresses assess more severe starting conditions. A proportional management cost gives an exact resource interpretation to a verified policy improvement under log utility. An independent scalar nonlinear HJB calculation evaluates learned feedback rules against a full-action numerical reference. Its domain and grid refinements make the quality of this low-dimensional reference observable; the high-dimensional continuous-time bound follows from its own theorem. Together, these exercises distinguish the value of a feasible learned correction, the incremental benefit of its construction method, and its remaining distance to the optimum.

1.1. *Relation to the literature*

Neural residual methods build on numerical dynamic programming and function approximation, not on a new principle of optimality. Projection and related methods exploit economic structure to approximate a Bellman equation; see [Judd \(1998\)](#). Derivative approximation by smooth neural networks provides useful representation results ([Hornik et al., 1990](#)), but does not establish that a particular finite architecture or optimizer solves a nonlinear boundary-value problem.

The differential critic is closely related to the neural PDE approach in [Sirignano and Spiliopoulos \(2018\)](#). Their DGM formulation includes differential and boundary conditions and treats HJB equations. The deep BSDE method of [Han et al. \(2018\)](#) also includes an HJB application. The distinguishing object here is the combination of an explicit feasible actor, block-specific evaluation, and economic error accounting across control and equilibrium problems. An analytical formula for the maximizing action is helpful when available, but is not a universal dividing line between neural PDE methods. Likewise, recursive dependence on the value is a nonlinear zeroth-order term; full nonlinearity in the PDE sense concerns the highest derivatives.

Recursive utility separates intertemporal substitution from risk aversion ([Epstein and Zin, 1989](#), [Duffie and Epstein, 1992](#)). Its infinite-horizon formulation requires particular care about utility-process selection and transversality; [Herdegen et al.](#)

(2021) explains why a stationary algebraic solution alone is insufficient. The time-inconsistency extension is an equilibrium problem in the sense of Björk et al. (2016), not an ordinary control problem with a relabelled flow payoff. Finally, stochastic approximation results require their own step-size, stability, and limiting-dynamics conditions (Borkar, 2024). They do not convert constant-step Adam into a convergence theorem for economic optimality.

The inherited continuous-capital result closes a different part of the error account: curvature and a verified coupling norm bound the gain from arbitrary adapted deviations, while a schedule-centred actor supplies a uniformly constrained neural policy. The resulting bound concerns the original diffusion rather than a nodal surrogate. Its stronger structural benchmark is reported alongside every fitted policy.

Section 2 states the controlled economy, Section 3 defines NBO, and Section 4 establishes the general verification accounts. Section 6 states the economic applications and locates their complete derivations. The capital results and Section 8 supply the absolute and policy-specific benchmarks. Sections 9–12 connect costates to welfare, specify observation and implementation, and report the executed comparisons. The supplement contains complete proofs, every primary and fixed-work comparison, reference and sensing diagnostics, and the full prior application records.

2. THE CONTROLLED ECONOMY

Let W be an m -dimensional Brownian motion on a filtered probability space satisfying the usual conditions. The state $S = (u, X)$ contains internal and external state variables. An action $a = (\theta, c, \pi, \dots)$ combines internal adjustment and external choices. The controlled state evolves as

$$dS_s = b(s, S_s, a_s) ds + \sigma(s, S_s, a_s) dW_s, \quad a_s \in \mathcal{A}(s, S_s). \quad (1)$$

The compact feasible correspondence $\mathcal{A}$ is part of the model. Actions are progressively measurable and must generate a well-defined state process satisfying the declared state restrictions. State viability does not follow from bounded actor outputs alone.

We work initially on a finite horizon T . A computational domain D is either invariant, equipped with a specified reflection mechanism, or stopped on exit. In the stopped case let $\tau = T \wedge \inf\{s \geq t : S_s \notin D\}$. The payoff $g(\tau, S_\tau)$ is specified on the entire parabolic stopping boundary, including the terminal face. For a fixed admissible policy a , utility is the policy-specific solution

$$Y_s^a = g(\tau, S_\tau) + \int_s^\tau f(v, S_v, a_v, Y_v^a) dv - \int_s^\tau Z_v^a dW_v. \quad (2)$$

Adjustment costs are included in f . For example, time-additive utility has $f(s, S, a, y) = r_0(s, S, a) - \rho y$, where r_0 includes all current resource and adjustment costs. Define $V^a(t, x) = Y_t^{a, t, x}$ and $V^*(t, x) = \sup_a V^a(t, x)$. The value inside a candidate policy's recursive objective is its own utility process, not the unknown optimum inserted into every candidate objective.

ASSUMPTION 1—Finite-horizon comparison class: *The controlled state problem is well posed for the admissible policies used below. The coefficients are continuous, locally Lipschitz in the state uniformly over actions, and bounded on the compact stopped domain. The payoff is continuous and bounded. On an invariant utility interval containing all policy values, f is continuous in its arguments and uniformly Lipschitz in utility with constant $L \geq 0$. The admissible class is stable under concatenation and has the measurable-selection properties needed for dynamic programming. The resulting HJB problem and the fixed-policy problems obey comparison in the bounded continuous class with the specified boundary conditions.*

The comparison requirement identifies both a function class and a boundary problem. It is not asserted merely because the primitive coefficients are continuous. On a bounded, regular Dirichlet domain, a boundary barrier is one way to verify attainment of the stopping payoff. Reflection requires the corresponding Neumann or oblique derivative condition and a well-posed reflected state process. Unbounded-state applications require growth bounds and localization. These conditions are checked separately from a neural residual.

For a smooth function v , write

$$\mathcal{L}^a v = b(t, x, a)^\top Dv + \frac{1}{2} \text{tr}\{\sigma(t, x, a)\sigma(t, x, a)^\top D^2 v\}, \quad (3)$$

$$\mathcal{H}^a[v] = f(t, x, a, v) + \mathcal{L}^a v, \quad F[v] = -v_t - \sup_{a \in \mathcal{A}(t, x)} \mathcal{H}^a[v]. \quad (4)$$

The value solves $F[V^*] = 0$ with the specified boundary data, in the viscosity sense where classical derivatives do not exist. The diffusion covariance is the positive semidefinite matrix $\sigma\sigma^\top$. No claim that the product of two arbitrary positive semidefinite matrices is symmetric positive semidefinite is needed.

A continuous martingale representation can also be used when its quadratic variation is Markov in an augmented state. If $d\langle M \rangle_s = Q(s, S_s, a_s)ds$, the covariance in (3) becomes $\sigma Q\sigma^\top$. A merely predictable process Q_s is not, without further state augmentation, a deterministic coefficient of a Markov HJB equation. Quadratic variation by itself also does not imply conditionally Gaussian increments. The implementation in this revision uses the Brownian model (1).

2.1. Infinite horizon and economic normalization

For a stationary problem, the boundary-value equation is $\sup_a \{f(x, a, V) + \mathcal{L}^a V\} = 0$. We use an infinite-horizon result only after specifying admissibility, integrability, and the selected utility process. In the time-additive case, a sufficient verification requirement is that the localized Itô identities are uniformly integrable and $\mathbb{E}[e^{-\rho T} | V(S_T) |] \rightarrow 0$ for the policies being compared. Recursive problems require an appropriate utility comparison and transversality condition instead of simply borrowing the additive one.

Cardinal transformations also require care. Multiplying a fixed utility function by a positive constant does not change its maximizer. Adding a function of an *endogenously controlled preference state* generally does. Consumption units, preference normalization, and terminal utility are therefore primitive economic choices in Section 32.

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The actor $a_\phi(t, x)$ and critic $v_\xi(t, x)$ represent policies and continuation values. An internal actor and an external actor can be separate parameter blocks; their joint action remains the input to the same economic generator. A smooth activation is

useful for exact differentiation, while a nonsmooth neural representation can be used with the positive-weight Bellman formulation below.

3.1. Feasibility and boundary data

For a box $[a, \bar{a}]$, a sigmoid map produces interior feasible actions but does not attain its endpoints at finite weights. Binding constraints can instead be represented by a projection onto the closed interval, by an explicit boundary candidate, or by a finite action set including the endpoints. The constrained portfolio calculation uses a projection and can attain a zero risky share exactly. On a simplex, a probability map supplies a feasible interior point and boundary faces must again be considered when certifying the optimum. General constraints require a feasible parameterization, a certified projection, or a constrained action solver. A finite penalty is not an exact feasibility certificate.

On a finite horizon, an elementary hard-terminal architecture is

$$v_{\xi}(t, x) = G(x) + (T - t)N_{\xi}(t, x). \quad (5)$$

This enforces the terminal face, but not automatically a lateral boundary. A compatible lifting G_B and a factor d_B vanishing on all Dirichlet faces give $v_{\xi} = G_B + d_B N_{\xi}$. Alternatively, boundary residuals are trained separately and retained explicitly in the error bound. A reflected preference state requires its own derivative condition; a terminal architecture does not enforce it. For recursive utility, a domain-preserving transformation must additionally keep the continuation value in the domain of the aggregator.

3.2. The block-specific computational graph

Let ν be a specified distribution of state–time points and let sg denote stop-gradient. A critic step minimizes

$$L_C(\xi; \bar{\phi}) = \mathbb{E}_{\nu} \left[-\partial_t v_{\xi} - \mathcal{H}^{\text{sg}(a_{\bar{\phi}})}[v_{\xi}] \right]^2 + L_B(\xi), \quad (6)$$

where L_B is absent only for boundary conditions enforced exactly. The policy, sampling locations, and any rival actions are detached. The value, time derivative, state gradient, and Hessian remain differentiable with respect to the critic parameters.

An actor step minimizes

$$L_A(\phi; \bar{\xi}) = -\mathbb{E}_v \mathcal{H}^{a\phi}[\text{sg}(v_{\bar{\xi}}), \text{sg}(Dv_{\bar{\xi}}), \text{sg}(D^2v_{\bar{\xi}})]. \quad (7)$$

All critic derivative channels are fixed in this step. There is no critic gradient from L_A and no actor gradient from L_C . A printed sum $L_A + \omega L_C$ may be a reporting device, but it is not jointly differentiated. The implementation exposes separate functions and regression tests for these derivative paths.

The distinction changes the population problem. With one action, $f(v) = -v$, $T = 1$, and terminal utility one, the correct solution is $v^*(t) = e^{t-1}$. The perturbation $v_\epsilon = v^* - \epsilon(1 - t)$ preserves the terminal condition. The joint objective used in the reviewed draft satisfies

$$\Delta(L_A + L_C) = -\frac{\epsilon}{2} + \frac{7\epsilon^2}{3}, \quad \Delta L_C = \frac{7\epsilon^2}{3}. \quad (8)$$

Thus the revised critic objective has the correct root as its minimizer along this perturbation, whereas the joint objective does not. The same problem is visible in a stationary equation: $-H + \omega H^2$ prefers $H = 1/(2\omega)$, not $H = 0$. No choice of finite positive weight resolves this identification error.

The actor objective is a local policy-improvement surrogate. Under differentiability of the policy value, the derivative of lifetime recursive utility from (t, x) instead involves the adjoint weight

$$D_\phi V^{a\phi}(t, x) = \mathbb{E}_{t,x} \int_t^\tau e^{\int_t^s f_y(v, S_v, a_v, V^{a\phi}) dv} D_a \mathcal{H}^{a\phi}[V^{a\phi}](s, S_s) D_\phi a_\phi(s, S_s) ds, \quad (9)$$

subject to the differentiability and integrability conditions given in the supplement. Arbitrary domain sampling does not reproduce this occupation and recursive-utility weighting. NBO uses (7) for approximate pointwise improvement and verifies the remaining action gap; it does not identify that loss with an exact lifetime policy gradient.

Algorithm 1 NBO evaluation, improvement, and acceptance

Require: Model, admissible actions, utility domain, boundary data, training and validation domains.

1: Initialize a feasible actor and a boundary-compatible critic.

2: **for** each outer iteration **do**

3: Freeze actor outputs. Update only the critic using (6) or a positive-weight Bellman evaluation residual.

4: Freeze the critic and every derivative channel. Improve only the actor using (7) or a feasible action solver.

5: Evaluate boundary error, policy-evaluation error, and the independent feasible-action gap.

6: Record the initialization, updates, conditioning, unsuccessful attempts, and stopping rule.

7: **if** the declared economic error criterion is met **then**

8: Return the policy, value, verification domain, and error report.

9: **end if**

10: **end for**

11: Return the computed approximation and its measured errors; do not infer optimality from optimizer stationarity.

3.3. Conditioning and stopping

A positive, parameter-independent state weighting can precondition the critic equation without changing its zero set. If that weight depends on learned quantities, its derivative must be included in the stated critic objective. The homothetic laboratory divides the Hamiltonian by its positive coefficient and differentiates that entire normalized critic residual. In the actor step the same coefficient is detached. Error conversion to the original economic equation includes the scale factor.

A zero residual target does not make learning dynamics stationary. For the scalar loss $\ell(z) = a^2 z^2 / 2$, gradient descent is stable only when $0 < \eta a^2 < 2$. Likewise, full differentiation of $\delta^2 / 2$ gives $\delta \nabla \delta$, while a semi-gradient TD update differentiates only the current-value term. The two computations must not be compared using

inconsistent signs or discount conventions. If f already includes $-\rho v$, a time discretization must not add a second independent discount to the same continuation equation.

Constant-step Adam is a numerical choice in the calibration laboratory, not an application of a two-timescale theorem. A genuine two-timescale analysis would require nonsummable and square-summable step sizes, an actor-to-critic ratio tending to zero, controlled noise, bounded iterates, and suitable attracting limiting dynamics (Borkar, 2024). A stationary critic gradient need not imply a zero residual, and a stationary multi-agent gradient need not imply Nash optimality. The results below therefore concern verified approximation errors rather than an assumed universal optimizer convergence property.

3.4. The implemented map and its verification output

Fix the economy, boundary realization, network classes, optimization budget, and verification procedure, denoted collectively by ϑ . Let $\mathcal{P}$ be the admissible feedback policies, Ξ the critic parameter space, and Ω the sampling streams. The implemented NBO is the numerical map

$$\mathcal{N}_\vartheta : \mathcal{P} \times \Xi \times \Omega \longrightarrow \mathcal{V}_\vartheta \times \mathcal{P}_\vartheta \times \mathcal{C}_\vartheta, \quad (a, \xi, \omega) \longmapsto (v, \tilde{a}, C). \quad (10)$$

The first component is approximate policy evaluation; the second is feasible improvement; the third records verification of the resulting pair. A record identifies the model, parameters, domain, boundary faces, evaluation error, maximization error, coverage, discretization errors, and unchecked assumptions. An unavailable bound is not zero. Failure to obtain a useful certificate is a possible output of the map. With its sampling stream fixed, the implementation is deterministic. We do not assert that a network learns an unrestricted Bellman operator on a function space.

The evaluation–improvement structure is an instance of approximate policy iteration, including its partial-evaluation variants (Scherrer et al., 2015). Neural representation does not introduce a new principle of optimality, and monotonicity belongs to the underlying positive-weight scheme, not to gradient descent. The

NBO specification joins the block-specific derivative graph to boundary and utility-domain restrictions and to a common economic error account. Sections 5 and 32.4 implement the differential and positive-weight versions of this map, respectively, with independent verification of their outputs.

3.5. Continuous-action proposals and a budget-safe Bellman enclosure

The learned map and its verifier have different computational roles. In the differential implementation, a feasible actor supplies an action and the critic evaluates that action's generator. In the positive-weight implementation, write

$$(T_n^a w)_i = r_{n,i}(a) + q_n \{P_{n,i}(a)w + b_{n,i}(a)\}, \quad (T_n w)_i = \sup_{a \in \mathcal{A}_i} (T_n^a w)_i. \quad (11)$$

Here $P_{n,i}(a)$ has nonnegative entries with sum at most one. The term b contains prescribed stopping payoffs, and $q_n > 0$. Continuous controls need not enter this expression concavely. Reflection, interpolation cells, and stopping decisions can change as the action changes. A supremum, rather than an attained maximizer, suffices below.

Our continuous-action implementation fits the actor by differentiating its own evaluated continuation payoff. It does not construct an all-action target or receive an exact maximizing label. A bounded tanh output enforces the control box. A local improvement step can subsequently compare that proposal with finitely many continuous perturbations. Such a step is recorded as part of a *hybrid* policy; it is not identified with the raw network and is not assumed globally optimal. Critic fitting uses the selected feasible action and the preceding continuation critic. Training ends before any exhaustive reference or action-cover verification begins.

For independent verification, freeze the complete policy $\pi = (\pi_n)$ and cover each action box by closed subboxes. At a state and time, let c_i be an outward lower bound obtained by evaluating any feasible action using a fixed continuation upper array. A box can be discarded only when its verified upper bound does not exceed $c_i + \tau$. Every box not so discarded is either subdivided or retained with its upper bound. Exhausting a time, depth, or box budget therefore enlarges the reported bound; it never converts an unresolved search into a success.

PROPOSITION 1—Budget-safe action and policy envelopes: Suppose the terminal arrays satisfy $L_N \leq g \leq U_N$. At each time n , evaluate the fixed-policy map outward to obtain $L_n \leq T_n^{\pi_n} L_{n+1}$. For every state, maintain a complete action cover whose box bounds enclose $T_n^a U_{n+1}$ for all actions in that box. Set $U_{n,i}$ to the maximum of the outward-rounded pruning threshold and all unresolved box upper bounds. Then, regardless of which boxes remain unresolved,

$$L_n \leq V_n^\pi \leq V_n^* \leq U_n, \quad 0 \leq V_n^* - V_n^\pi \leq U_n - L_n. \quad (12)$$

All inequalities are on the stated nodal economy. No concavity, exact training labels, or convergence of neural optimization is required. If $U_n - T_n U_{n+1} \leq e_n \mathbf{1}$, then its optimal-value excess is at most

$$\left(\prod_{j=n}^{N-1} q_j \right) \|U_N - g\|_\infty + \sum_{k=n}^{N-1} \left(\prod_{j=n}^{k-1} q_j \right) e_k.$$

The proof, including stopping faces, is in the supplement. A stored unresolved bracket supplies an upper bound for e_n ; it is not replaced by the requested tolerance. The policy lower envelope and the Bellman upper envelope are computed separately. The latter may be shared across several frozen policies on the same economy, while each policy requires its own evaluation. This separation prevents an expensive verification oracle from becoming undisclosed actor supervision.

In the preference application, the continuous box has three coordinates and the state grid has two. Bilinear interpolation is enclosed cell by cell. On each rectangle, its extrema occur at cell corners; a second implementation uses cell-slope enclosures. If a box crosses a wealth stopping face, the verifier takes the union of the interior continuation and the prescribed exit payoff. Their numerical values need not coincide at off-grid exit locations. Treating the entire action objective as globally differentiable at this switch would invalidate a derivative-only bound. Reflection is handled by including every crossed fold. Positive consumption and a preference interval strictly above one keep the powers and denominators in their stated domains.

The complete-action bound is not a continuous-state or continuous-time theorem. The verifier interprets stored primitive coefficients and nodal abscissae as exact

real inputs and encloses subsequent operations outward. Training and verification agree to numerical precision on the inherited action grid, but the certificate's precise object is this declared nodal economy, not an unspecified sequence of floating-point evaluations. The supplemental transfer result states separately the state interpolation, diffusion approximation, boundary monitoring, and action errors needed to recover the continuous economy. None is inferred from a sampled residual or silently assigned zero.

4. VERIFICATION, APPROXIMATION, AND VISCOSITY SELECTION

For a feasible policy a and a smooth candidate v , define

$$r^a[v] = -v_t - \mathcal{H}^a[v], \quad g^a[v] = \sup_{b \in \mathcal{A}} \mathcal{H}^b[v] - \mathcal{H}^a[v] \geq 0. \quad (13)$$

The first error concerns evaluation; the second concerns improvement. Their difference is the maximized HJB residual: $F[v] = r^a[v] - g^a[v]$. We keep the two quantities separate because the policy value and its regret depend differently on them.

THEOREM 2—Finite-horizon economic error bound: *Under Assumption 1, suppose a bounded smooth candidate v in the declared utility interval and an admissible Markov policy a satisfy $|r^a[v]| \leq e$, $0 \leq g^a[v] \leq \eta$ throughout the verification domain. Suppose the boundary payoff error is at most B . Any reflected boundary condition is imposed exactly. Set $A_L(s) = (e^{Ls} - 1)/L$ for $L > 0$ and $A_0(s) = s$. Then, for $s = T - t$,*

$$|v(t, x) - V^a(t, x)| \leq e^{Ls}B + A_L(s)e, \quad (14)$$

$$|v(t, x) - V^*(t, x)| \leq e^{Ls}B + A_L(s)(e + \eta), \quad (15)$$

$$0 \leq V^*(t, x) - V^a(t, x) \leq 2e^{Ls}B + A_L(s)(2e + \eta). \quad (16)$$

Consequently, exact evaluation, exact boundary data, and a zero feasible-action gap identify the optimal value and an optimal policy.

The theorem replaces an assertion about a global minimizer of a composite loss with a statement about the exact computational objects NBO estimates. It allows errors, does not require that a finite network class contain V^* , and bounds performance against all admissible policies covered by comparison. A small sampled

residual is not yet the uniform hypothesis of the theorem. Section 4.3 addresses that distinction.

4.1. A positive-weight recursive Bellman approximation

Let P_h^a be a nonnegative transition/interpolation operator on a finite state set, with row sums one after adjoining absorbing states. Boundary values are fixed. For a continuation array w , define $\Psi_h^a w$ pointwise as the scalar solution

$$z = P_h^a w + hf(x, a, z), \quad T_h w = \max_{a \in \mathcal{A}_h(x)} \Psi_h^a w. \quad (17)$$

For stationary calculations assume $f_y \leq -\lambda < 0$ on an invariant value domain, and require that all scalar roots remain in that domain. The map $z \mapsto z - hf(x, a, z)$ is strictly increasing with slope at least $1 + \lambda h$. Hence the root is unique and Ψ_h^a is order preserving and a sup-norm contraction with factor $q_h = (1 + \lambda h)^{-1}$. The maximum over feasible actions preserves these properties. A root bracket must contain the root; arbitrary clipping of utility values is not part of the definition.

For a finite horizon, the same implicit construction is used backward from the prescribed terminal array. Under the Lipschitz condition in Assumption 1, $hL < 1$ gives a one-step Lipschitz constant at most $(1 - hL)^{-1}$. Discount contraction is not assumed for a recursive parameter regime where it has not been verified. For a time-additive problem, the positive-weight backup $hr_0 + e^{-\rho h} P_h^a w$ is also consistent and monotone. It is the backup used by the NDU reference laboratory.

THEOREM 3—Discrete NBO certificate: *Suppose (17) is well defined on an invariant domain, with contraction factor $q_h < 1$. Let v_h^* and v_h^a denote the fixed points of T_h and T_h^a , respectively. For any neural value array v and deterministic feasible policy a , set*

$$\delta = \|v - T_h^a v\|_\infty, \quad \epsilon = \|T_h v - T_h^a v\|_\infty.$$

Then

$$\|v - v_h^a\|_\infty \leq \frac{\delta}{1 - q_h}, \quad \|v - v_h^*\|_\infty \leq \frac{\delta + \epsilon}{1 - q_h}, \quad \|v_h^* - v_h^a\|_\infty \leq \frac{2\delta + \epsilon}{1 - q_h}. \quad (18)$$

The certificate covers the stated finite state and action model. Continuous actions require an additional certified maximization or action-coverage error.

This result applies regardless of how the neural array was obtained. Neural projection need not itself be monotone. What matters is that the output satisfies the monotone Bellman equation to the displayed tolerance. The distinction makes it possible to use flexible representations while retaining a comparison-based acceptance test.

THEOREM 4—Viscosity-consistent neural Bellman approximation: *Consider stationary problems satisfying comparison in a bounded continuous class and the strict utility monotonicity used in Theorem 3. Suppose the positive-weight schemes are stable, locally consistent with the HJB operator, and attain the prescribed boundary data in the limit. The feasible action approximations become dense with vanishing maximization error. If neural arrays and feasible actors satisfy*

$$\delta_h + \epsilon_h = o(h), \quad (19)$$

then their interpolated values converge locally uniformly to the unique viscosity solution. Uniform convergence up to the boundary follows when boundary barriers give uniform boundary attainment. Their discrete policy-value arrays converge to the same limit. No assertion of convergence of the policy functions themselves is required.

Here stability, local consistency, comparison, and boundary attainment are numerical-analysis hypotheses, not consequences of calling a representation neural. The proof is given in the appendix. A finite-horizon version uses backward error accumulation: terminal error tends to zero and the sum of one-step errors tends to zero after multiplication by the one-step Lipschitz factors. This covers finite-horizon recursive problems without imposing stationary discount contraction.

For a diffusion with m Brownian factors, an interior semi-Lagrangian construction uses the $2m$ locations $x + hb \pm \sqrt{mh} \sigma e_j$, each with weight $1/(2m)$. Positive multilinear interpolation preserves monotonicity. For smooth test functions its interpolation contribution is $O(\Delta x^2/h)$; thus $\Delta x^2/h \rightarrow 0$ is a sufficient interior coupling condition. The mean and covariance of the cubature increments match those of the specified Brownian diffusion to first order. Stopping and reflection require consistent boundary transitions. Negative interpolation weights are not permitted in this construction.

4.2. Why an integrated differential residual does not select the value

Consider the exit problem $F(v') = 1 - |v'| = 0$ on $(-1, 1)$ with zero boundary payoff. Its value is $v^*(x) = |x| - 1$. The smooth functions

$$w_\epsilon(x) = \sqrt{1 + \epsilon^2} - \sqrt{x^2 + \epsilon^2}$$

satisfy the same boundary values and have vanishing integrated squared residual, but converge uniformly to $1 - |x|$. At zero the test function one touches the wrong limit from above and gives $F(0) = 1 > 0$, violating the viscosity subsolution inequality. The defect is not removed by exact Hamiltonian maximization.

On a grid with spacing h , the positive-weight exit backup is

$$(Tv)_i = -h + \max(v_{i-1}, v_{i+1}), \quad v_0 = v_N = 0. \quad (20)$$

It is an undiscounted shortest-exit problem, not an instance of the stationary contraction theorem. Backward propagation of the boundary values gives $v_i = |x_i| - 1$ exactly. At the peak of the wrong candidate, the *scaled* residual $|w_i - (Tw)_i|/h$ is close to two, rather than close to zero. The laboratory therefore distinguishes correct selection from small integrated differential error.

An explicit positive viscosity term is another possible regularization, provided comparison, boundary convergence, and approximation errors for the regularized equations are controlled before taking the viscosity to zero. Network width or spectral bias is not itself a positive viscosity parameter. Moreover, approximation of a C^2 function and its derivatives does not establish approximation of nonexistent classical derivatives at a kink.

4.3. Coverage and independent maximization

A verification distribution is declared independently of the actor's occupation measure. It includes economically relevant counterfactual states and boundary regions, not only states visited by a current policy. If a residual r has a known Lipschitz bound K on a compact domain and a validation set has fill distance d_{fill} ,

then

$$\|r\|_\infty \leq \max_{x \in \mathcal{V}} |r(x)| + Kd_{\text{fill}}. \quad (21)$$

The same bound can be applied to an action gap if its modulus is controlled. An empirical mean alone does not give this guarantee. If the sampling density is bounded below and the domain has a uniform interior-volume condition, a Lipschitz residual also admits an L^2 -to-sup estimate with exponent $1/(d+2)$, as derived in the supplement. Its dimension dependence is material.

An independent action solver must provide an *upper bound* on the unobserved maximization error. Trying a few alternative actions supplies only a lower bound on the true gap. Concavity with a duality gap, interval bounds, exhaustive enumeration of a finite action set, or an action net with a known Lipschitz modulus can provide the needed upper bound. An unverified local actor optimizer cannot certify itself.

4.4. Differential and discrete error accounts

For the stationary additive problem, write $f = r_0 - \rho v$, with $\rho > 0$, and let R_h restrict a candidate to the nodes of a positive-weight scheme. On interior nodes set

$$T_h^a w = q_h(P_h^a w + h r_0^a), \quad q_h = (1 + \rho h)^{-1}.$$

Stopping nodes instead use their prescribed payoff.

PROPOSITION 5—Transfer to a Bellman certificate: *Suppose the represented policy is feasible for the discrete action class and*

$$\sup_a \left\| h^{-1}(P_h^a R_h v - R_h v) - R_h \mathcal{L}^a v \right\|_{\infty, \text{int}} \leq \kappa_h.$$

If the differential evaluation residual is at most e and its feasible-action gap at most η , then, on the interior nodes,

$$\|R_h v - T_h^a R_h v\|_\infty \leq q_h h(e + \kappa_h), \quad \|T_h R_h v - T_h^a R_h v\|_\infty \leq q_h h(\eta + 2\kappa_h).$$

Boundary discrepancies and finite-action coverage errors are additional accounts.

The proof follows by adding and subtracting $R_h \mathcal{L}^a v$; the maximization inequality uses the consistency error twice. The converse is false for arbitrary oscillatory interpolants. For smooth interior functions, local cubature and interpolation estimates can bound κ_h ; they do not justify assigning zero error to a stopping or reflected boundary without checking its realization.

For finite-horizon maps with Lipschitz factors β_t , let δ_t bound evaluation error, ϵ_t bound improvement error, and B_N bound terminal error. Define

$$E_N = B_N, \quad O_N = B_N, \quad R_N = 2B_N, \quad \begin{cases} E_t = \delta_t + \beta_t E_{t+1}, \\ O_t = \delta_t + \epsilon_t + \beta_t O_{t+1}, \\ R_t = 2\delta_t + \epsilon_t + \beta_t R_{t+1}. \end{cases} \quad (22)$$

Backward comparison bounds policy-value error by E_t , optimal-value error by O_t , and policy regret by R_t . These arrays are computed, rather than replaced by a training loss, in the NDU experiment. Bounds at later time indices must use their own accounts, not necessarily the account at time zero. Representation and incomplete optimization enter δ_t ; action coverage enters ϵ_t ; the error of replacing the controlled diffusion by a scheme is separate. If a uniform continuous-to-discrete policy-payoff discrepancy is bounded by D_h , transferring a discrete regret bound adds $2D_h$. Without that estimate, the certificate remains a certificate for the stated discrete economy. In particular, this accounting does not establish the optimizer rate required by Theorem 4.

5. A TRAINED NEURAL BOUNDARY-VALUE PROBLEM

Consider a household with labor income, risky wealth, and a prescribed retirement payoff on exit. Before exit from $D = (0.5, 2.5)$,

$$dX_s = (0.12 + 0.02X_s - c_s) ds + 0.15X_s dW_s, \quad c_s \in [0.02, 0.6]. \quad (23)$$

Discount is $\rho = 0.08$, flow utility is $\log c$, and exit pays $g(x) = \log(0.12 + 0.08x)/0.08$. This is a specified stopped economy, not a claim about an unbounded-wealth problem. Income and the exit option destroy the homothetic reduction used in the portfolio calibration. Bounded coefficients, a compact positive consumption

interval, positive diffusion variance on the closed domain, and continuous endpoint data yield the usual stopped-domain comparison problem. A neural approximation must solve its nonlinear differential equation and both boundary conditions.

The critic has two hidden tanh layers of width sixteen and the lifting

$$v_{\xi}(x) = g(\ell) + \frac{g(u) - g(\ell)}{u - \ell}(x - \ell) + (x - \ell)(u - x)N_{\xi}\left(2\frac{x - \ell}{u - \ell} - 1\right).$$

The actor uses the same hidden architecture and a sigmoid into the consumption interval. Here the optimum is not forced to an endpoint by that representation. Independent verification nevertheless maximizes over the *closed* interval: for $p = v'(x)$,

$$c^*(p) = \begin{cases} \text{clip}(1/p, 0.02, 0.6), & p > 0, \\ 0.6, & p \leq 0. \end{cases} \quad (24)$$

Training imports the general differential core used for the earlier routing tests. It does not fit reference values, derivatives, or policy labels.

PROPOSITION 6—Continuous-domain economic certificate: *For a stopped discounted additive economy obeying comparison, let $v \in C^2(D) \cap C(\overline{D})$ and let a be admissible. Suppose throughout D that*

$$|\rho v - r_0^a - \mathcal{L}^a v| \leq e, \quad 0 \leq \sup_b (r_0^b + \mathcal{L}^b v) - (r_0^a + \mathcal{L}^a v) \leq \eta, \quad \|v - g\|_{\partial D} \leq B.$$

Then

$$\|v - V^a\|_{\infty} \leq B + \frac{e}{\rho}, \quad \|v - V^*\|_{\infty} \leq B + \frac{e + \eta}{\rho}, \quad 0 \leq V^* - V^a \leq 2B + \frac{2e + \eta}{\rho}. \quad (25)$$

Indeed, constant upper and lower shifts of v dominate the evaluation or optimality residual and the stopping data; comparison gives the first two inequalities and their sum gives the third. No reference-grid error is used in this argument.

The executable verifier covers the entire state interval by dyadic subintervals. Outward-rounded propagation through the stored network encloses v , v' , v'' , and the actor on every leaf. Equation (24) supplies a global action enclosure. We also

use the concavity bound

$$\log c^* - pc^* - (\log c - pc) \leq \frac{1}{2}C^2(1/c - p)^2, \quad C \geq \max\{c, c^*\}, \quad (26)$$

which follows by completing the square in the strong-concavity inequality on the segment joining the two actions. It does not assume an interior maximizer. Taking the better of this bound and a direct interval enclosure avoids dependency inflation. Terminal-free boundary errors are evaluated separately at both endpoints, including rounding in the lifting constants.

The certificate saves every leaf, verifies that their union is exactly $[0.5, 2.5]$, and binds the parameter and arithmetic files by hashes. Exponential and logarithmic enclosures use range reduction and polynomial remainder bounds rather than an assumed accuracy of a library transcendental. Its mathematical conclusion is conditional on the stated binary64 arithmetic contract; high-precision and derivative tests check the implementation but are not a formal machine proof. The construction does not estimate a global Lipschitz constant from samples.

5.1. *Controlled comparison and failures*

We use seeds 11, 29, and 47, each with 256 prespecified jittered collocation points. Eight outer rounds have a common cap of 800 L-BFGS inner iterations. NBO allocates each round sixty iterations to evaluation and forty to improvement; direct neural HJB uses the same critic and the global action rule in all one hundred iterations. A joint-loss ablation differentiates the sum of squared residual and negative Hamiltonian through both blocks. Line searches use different numbers of closures, so this is a matched iteration cap, not an assertion of equal floating-point work. Closure counts and elapsed times are saved.

An independent positive-rate upwind policy-iteration solver uses continuous-action maximization within each drift region. Reference grids have 320, 640, 1,280, and 2,560 intervals. Value and derivative comparisons use the finest grid; its refinement differences are not substituted for a rigorous discretization bound. The continuous certificate instead checks the neural functions directly. All three NBO and direct-HJB runs attain the declared interval-verification thresholds. The joint-

TABLE I

MULTILAYER CONSUMPTION: APPROXIMATION AND CONTINUOUS-DOMAIN VERIFICATION

Method	Seed	$\ v - V_h^*\ _\infty$	Sampled e	Uniform η	Regret bound
NBO	11	3.30×10^{-4}	6.18×10^{-4}	4.24×10^{-6}	0.0501
NBO	29	5.75×10^{-4}	0.0015	2.66×10^{-6}	0.0500
NBO	47	5.46×10^{-4}	0.0014	4.36×10^{-6}	0.0501
Direct HJB	11	4.14×10^{-4}	9.72×10^{-4}	0	0.0500
Direct HJB	29	3.55×10^{-4}	7.13×10^{-4}	0	0.0500
Direct HJB	47	6.00×10^{-4}	0.0017	0	0.0500
Joint loss	11	5.4142	0.5002	—	—
Joint loss	29	5.4435	0.5003	—	—
Joint loss	47	5.5677	0.5003	—	—

The comparison grid has 2,560 intervals on $[.5, 2.5]$. The last two columns cover the entire continuous interval and consumption set $[.02, .6]$. All six certificates have uniform evaluation error at most .002 and endpoint error below 10^{-12} . Dashes indicate no accepted certificate, not zero error. Reference error and the sampled differential residual are not used as uniform bounds.

loss ablation does not: a residual near one half is accompanied by large value and action errors. Width, depth, and update-allocation variations are retained in the supplement, including unfavorable outcomes. The direct method is a strong comparator: these results do not establish that carrying an actor is uniformly faster or more accurate. They establish an end-to-end neural solve with an independently computable welfare bound, and identify a concrete failure of joint differentiation. Fitting and full-domain verification are separately timed because their computational burdens are different.

6. ECONOMIC APPLICATIONS OF THE BELLMAN STRUCTURE

The motivating economic problem combines external choices with the costly adjustment of preferences. Recursive utility changes how a given policy is evaluated; preference adjustment adds an internal choice to its Hamiltonian; and strategic interaction changes whose deviations define improvement. These distinctions determine the economic interpretation of NBO. The capital economy studied below provides a setting in which policy-specific verification can be carried through to a

high-dimensional diffusion. The preference, temporal-self, and game applications establish the corresponding economic objects and their own verification results.

6.1. Costly preference adjustment and portfolio choice

Let u denote risk aversion and X financial wealth. The household chooses consumption c , portfolio share p , and preference-adjustment effort θ . The preference state is reflected on a declared interval $[\underline{u}, \bar{u}] \subset (1, \infty)$, and wealth follows the portfolio diffusion:

$$\begin{aligned} du &= \theta dt + \sigma_u dW^u + dL^u - dL^{\bar{u}}, \\ dX &= \{[r + p(\mu - r)]X - c\} dt + p\sigma_X X dW^X, \quad d\langle W^u, W^X \rangle_t = \varrho dt. \end{aligned} \quad (27)$$

For a fixed consumption unit $\bar{c} > 0$, flow utility is

$$U(c, u) = \frac{1}{2}k\theta^2, \quad U(c, u) = \frac{(c/\bar{c})^{1-u}}{1-u}.$$

The objective discounts both felicity and adjustment costs at rate ρ and includes the specified wealth-exit and terminal payoff. The baseline uses $\bar{c} = 1$. Because the household controls u , the cardinal normalization across preference states is a primitive: adding a function of u changes the incentives to adjust preferences. The reflected state, positive consumption domain, and stopping data give the evaluation equation its economic and mathematical domain.

At an interior optimum with $V_X > 0$ and $V_{XX} < 0$, the three policy conditions are

$$\begin{aligned} c^* &= \bar{c}(\bar{c}V_X)^{-1/u}, & \theta^* &= V_u/k, \\ p^* &= -\frac{V_X(\mu - r)}{XV_{XX}\sigma_X^2} - \frac{\varrho\sigma_u V_{uX}}{XV_{XX}\sigma_X}. \end{aligned} \quad (28)$$

Box constraints require their corresponding feasible maximizers. The cross derivative in the portfolio rule links the external risk choice to the internal preference state. Its sign is an equilibrium object to be determined by evaluation. The adjustment rule likewise relates effort to the shadow value of changing preferences. A higher adjustment cost lowers the value of every fixed policy and hence the optimized

value; whenever an envelope derivative exists,

$$\partial_k V^k(t, u, X) = -\frac{1}{2} \mathbb{E}_{t,u,X}^{a^k} \int_t^\tau e^{-\rho(s-t)} \theta_s^2 ds \leq 0. \quad (29)$$

This comparative static holds with the feasible class and stopping payoff fixed.

The numerical application implements these choices jointly. The finite-action study evaluates complete policies by independent backward recursion. The continuous-action study trains the actor through its own payoff and verifies its hybrid policy with a complete action-box enclosure. Fresh training on nested grids separates the effect of resolution from the evaluation of a frozen coarse policy. A model-specific compensating-consumption calculation translates its cardinal utility bounds into an explicit transfer experiment. Supplementary Sections 32, 32.4, and 37 give the full model, all recorded policies, refinement evidence, and compensation account.

6.2. Recursive utility and temporal selves

Recursive utility enters policy evaluation through the policy's own continuation utility. With $a = 1 - 1/\psi$, $b = 1 - \gamma$, and $bV > 0$, the Epstein–Zin aggregator used here is

$$f(c, V) = \frac{\rho}{a} \{c^a (bV)^{1-a/b} - bV\}. \quad (30)$$

The invariant utility interval and boundary payoff are part of the evaluation problem. Supplementary Section 31 derives the homothetic policy conditions and verifies the matching-bequest experiment, including the barriers that keep every policy utility in the aggregator's domain. This application tests recursive evaluation while maintaining the same separation between value approximation and feasible action improvement.

Time inconsistency requires a different improvement criterion. For the continuous discount kernel

$$h(s) = \beta e^{-\rho s} + (1 - \beta) e^{-(\rho + \kappa)s}, \quad 0 < \beta \leq 1, \quad \kappa > 0,$$

two critics evaluate the same future policy at the two discount rates. Writing these values as W_1, W_2 , the current self uses $V = \beta W_1 + (1 - \beta) W_2$ and chooses an in-

stantaneous deviation by maximizing $u(a) + \mathcal{L}^a V$. Supplementary Section 33 states the complete component equations and derives the spike-deviation condition. In the homothetic consumption economy, the equilibrium consumption share is the unique root in its admissible interval, and stronger present bias raises that share. Independently optimized finite-duration deviations check the limiting equilibrium condition. Thus the actor's objective follows from the temporal equilibrium concept, while the critics retain their policy-evaluation role.

6.3. Capital investment and strategic deviations

In the dynamic Cournot application, firm i chooses investment in its own capital, with

$$dK^i = (I^i - \delta K^i) dt + \sigma_K K^i dW^i, \quad q_i = AK^i.$$

Its flow profit is $P_N(K)q_i - I^i - b_i(I^i)^2/2$, where the market-size normalization in P_N is specified for each economy. Given rivals' policies, a critic evaluates the resulting own-firm payoff and the actor changes only its own action. Consequently, the relevant economic error is the gain from a complete unilateral deviation against the frozen rival profile.

Supplementary Sections 34 and 34.3 specify the boundaries, investment sets, and market normalizations, and report independent full dynamic best responses to every principal finite neural profile. The resulting profit-unit bounds include all dates and states of the stated finite game. They also cover randomized and history-dependent unilateral deviations against those fixed Markov rivals, by finite dynamic programming. The record reports the cost of the exhaustive training oracle and any nonzero one-step equilibrium defects along with the final exploitability calculation.

These applications give the common method a precise economic interpretation. A recursive critic evaluates a utility process; preference and capital actors allocate resources and adjustment effort; a temporal actor satisfies a spike condition; and a strategic actor is assessed against unilateral best responses. Each application supplies the boundary conditions, approximation class, and error account required by its own economic question.

7. CONTINUOUS-ECONOMY GUARANTEES FOR NEURAL CAPITAL POLICIES

The distinction between a nodal certificate and an economic diffusion bound is substantive. This section obtains the latter for the original dense, nonlinear capital economy, without replacing its primitives, restricting the comparison policy class, or extrapolating a sampled residual. The construction uses the curvature of consumption utility to control the gain from an arbitrary adapted deviation. A neural actor is then confined to a known neighbourhood of a feasible schedule. Its approximation class is restricted; the economic optimum against which it is assessed is not.

Write $\bar{z} = d^{-1} \sum_i z_i$, $\|z\|_d^2 = d^{-1} \sum_i z_i^2$, and $\Pi z = z - \bar{z} \mathbf{1}$. In the capital model already specified, the state and payoff are

$$dY_i = \{a + \kappa \tanh((BY)_i) - m_i - \nu^2/2\} dt + \sigma_I dW_i + \sigma_C dW_0, \quad (31)$$

$$J_y(m) = \mathbb{E}_y \left[\int_0^T e^{-\rho t} \{\overline{\log m_t} + \bar{Y}_t - \frac{\zeta}{2} \bar{m}_t^2\} dt + e^{-\rho T} \{\bar{Y}_T - \chi \|\Pi Y_T\|_d^2\} \right], \quad (32)$$

where $\nu^2 = \sigma_I^2 + \sigma_C^2$ and each control lies in $[\underline{m}, M]$. Controls are progressively measurable and the state equation is interpreted on $\mathbb{R}^d$. Bounded control and the globally Lipschitz bounded production term yield the required state moments and finite payoffs. The historical parameters are $T = 1$, $\rho = .04$, $a = .10$, $\kappa = .15$, $\sigma_I = .15$, $\sigma_C = .10$, $\zeta = .2$, $\chi = .03$, and $[\underline{m}, M] = [.02, 2]$. The stored coupling matrices are unchanged in dimensions ten and twenty. Dimension fifty uses the same stated matrix-generation rule.

Define

$$w(t) = \frac{1 - e^{-\rho(T-t)}}{\rho} + e^{-\rho(T-t)}, \quad m_0(t) = \frac{2}{w(t) + \sqrt{w(t)^2 + 4\zeta}}. \quad (33)$$

This schedule is strictly interior for the parameters above. It is a feasible benchmark, not an asserted solution of the coupled economy. Let $\beta \geq \kappa \|B\|_2$, let s_0 bound the initial cross-sectional standard deviation, and set

$$Q_d = s_0 + \kappa T + \sigma_I \sqrt{(1 - 1/d)T}, \quad (34)$$

$$K(s) = \beta \int_s^T e^{-\rho t} w(t) e^{\beta(t-s)} dt + 2\chi e^{-\rho T} Q_d e^{\beta(T-s)}, \quad (35)$$

$$\mu(t) = \frac{2\{\log[m_0(t)/M] + [M - m_0(t)]/m_0(t)\}}{[M - m_0(t)]^2}. \quad (36)$$

The quantity $\mu(t)$ is the smallest tangent-remainder curvature of log utility on the entire primitive action interval, relative to $m_0(t)$. It is not a Hessian bound estimated from training points. In particular, it can be substantially sharper than the uniform lower curvature M^{-2} .

THEOREM 7—Curvature-controlled policy guarantee: *Suppose $\rho > 0$, $\zeta, \chi, \kappa \geq 0$, the schedule in (33) is interior, and $\|\Pi y\|_d \leq s_0$. Let $V^*(y) = \sup_m J_y(m)$, where the supremum is over the full original admissible class. Then*

$$0 \leq V^*(y) - J_y(m_0) \leq G_d := \frac{1}{2} \int_0^T e^{\rho s} \frac{K(s)^2}{\mu(s)} ds. \quad (37)$$

For $\epsilon \geq 0$ satisfying $[m_0(t) - \epsilon, m_0(t) + \epsilon] \subset [\underline{m}, M]$, every admissible policy m^ϕ with $|m_i^\phi(t, Y) - m_0(t)| \leq \epsilon$ for all coordinates, times, and states satisfies

$$0 \leq V^*(y) - J_y(m^\phi) \leq G_d + D_{d,\epsilon}, \quad (38)$$

where, with $A_\beta(T) = \int_0^T e^{\beta t} dt$,

$$D_{d,\epsilon} = \int_0^T \left[\epsilon K(s) + \frac{\epsilon^2 e^{-\rho s}}{2} \{[m_0(s) - \epsilon]^{-2} + \zeta\} \right] ds + \chi e^{-\rho T} \epsilon^2 A_\beta(T)^2. \quad (39)$$

These bounds hold for every initial mean, every initial state with the stated dispersion bound, and arbitrary adapted competitors in the original action box. They require no neural training convergence or discretization error estimate.

The proof uses synchronous state comparison, an exact discounted mean-state identity, and completion of a quadratic bound on an arbitrary policy deviation. It is provided in the appendix. The result is a model-specific numerical principle: curvature limits the benefit from changing consumption, whereas a spectral bound

limits the propagation of that change through dense nonlinear production. Their balance yields a scalar integral certificate for a high-dimensional policy. Bounded nonlinearity, the additive diffusion structure, and the chosen felicity are essential assumptions, rather than generic properties of all HJB equations.

7.1. The deployed neural map and its numerical guard

The actor is a two-hidden-layer network with the closed-form output map

$$m_i^\phi(t, Y) = m_0(t) + \epsilon \tanh N_{\phi,i}(t, Y). \quad (40)$$

The weights are trained by the same block-specific NBO losses as before. No maximizing policy labels, nodal action table, or online local search enter this actor. Input clipping at very large log-capital values protects numerical evaluation; it does not stop or truncate the economic state. The theorem remains valid for any bounded proposal followed by the stated output constraint, including this clipped-input network.

Numerical implementation must not replace the exact schedule by an unverified floating-point centre. The deployment routine therefore encloses $m_0(t)$ by $[\ell(t), u(t)]$ using the inherited outward arithmetic kernel and projects the proposal onto

$$[\text{up}(u(t) - \epsilon), \text{down}(\ell(t) + \epsilon)]. \quad (41)$$

This interval lies inside the exact tube. Adjacent clock floats enclose rounding of the input time. An empty interval causes an error; a nonfinite proposal is replaced by its interval midpoint and is not evidence of successful training. This scalar feasibility guard is neither an action-maximization oracle nor a stored policy correction table. Its cost is included in the deployment measurements.

7.2. Computable scope and cost

An SVD is used only to propose a scalar λ . Positive pivots in an outward interval $LDL^\top$ calculation for $\lambda I - B^\top B$ certify the spectral majorant. A failed pivot triggers enlargement, not acceptance of the SVD output. The remaining integrals use outward rectangle sums, polynomial exponential and logarithm bounds, and

square-root endpoints checked by interval squaring. For a fixed majorant with verified pivots, this certificate requires $O(d^3)$ arithmetic operations and $O(d^2)$ storage for the matrix check, followed by $O(N)$ scalar quadrature work. This is not a dimension-independent training theorem or a worst-case bound on the number of attempted majorants.

The guarantee holds already at initialization. Training is intended to improve the feasible policy inside the tube, not to manufacture its certificate. The anchor itself has the tighter bound G_d and requires no training. A comparison omitting it would overstate the role of the actor. Neither a small $G_d + D_{d,\epsilon}$ nor faster actor evaluation proves a net economic advantage over that benchmark.

8. POLICY-SPECIFIC VERIFICATION AFTER BELLMAN IMPROVEMENT

The bound in Theorem 7 is a useful starting point, but not a measure of what training accomplishes. Every policy in the same tube receives the same allowance, including a poorly trained policy. We now replace that allowance by an independently verified, policy-specific payoff difference. The economic primitives, initial state, and full adapted comparison class in (31)–(32) are unchanged. The implementation to which the new bound applies is specified explicitly below. It is not identified with continuously observed-state feedback.

8.1. The deployed controller

Fix a deterministic grid $t_k = kh$, $Nh = T$. The controller has an internal state Z_k , initially y , and observes the Brownian innovations at the grid dates. At date t_k it selects

$$a_k = \mathcal{P}_k\{m_\phi(t_k, Z_k)\}, \quad Z_{k+1} = Z_k + h\{c\mathbf{1} + f(Z_k) - a_k\} + \Sigma\Delta\tilde{W}_k, \quad (42)$$

where $c = a - v^2/2$, $f(z) = \kappa \tanh(Bz)$, and $\mathcal{P}_k$ projects inward onto the declared action interval around $m_0(t_k)$. The vector $\Delta\tilde{W}_k$ clips each standardized innovation at ten before rescaling by $\sqrt{h}$. The numerical implementation and its rounding convention form part of the controller. Between dates it holds a_k fixed. Its economic

state, denoted Y^ϕ , nevertheless satisfies

$$dY_t^\phi = \{c\mathbf{1} + f(Y_t^\phi) - a_k\} dt + \Sigma dW_t, \quad t_k \leq t < t_{k+1}.$$

In particular, the economic Brownian motion and state are not clipped, stopped, or replaced by Z . The controller is adapted to the original innovation filtration and remains feasible. Observing innovations is an implementation assumption, not a conclusion inferred from a state-only observation model. The continuously evaluated R10 policies remain separate objects in the supplement.

This specification has an important analytical consequence. When the numerical state and the economic state are compared, they use exactly the same chosen actions. Their state discrepancy therefore depends on the smoothness of the economic drift, not on a global Lipschitz bound for the learned network. The following construction uses this fact rather than inserting sampled residuals into a uniform verification theorem.

8.2. An exact improvement identity

Write $W(t) = e^{-\rho t}w(t)$ and define the nonnegative consumption deficit

$$R_t(m) = \log m_0(t) - \overline{\log m} + w(t)\{\bar{m} - m_0(t)\} + \frac{\zeta}{2}\{\bar{m}^2 - m_0(t)^2\}. \quad (43)$$

The first-order identity $m_0^{-1} = w + \zeta m_0$ and concavity of the logarithm imply $R_t(m) \geq 0$. The exact mean-state identity in the proof of Theorem 7 gives

$$\begin{aligned} \Delta_\phi := J_y(a^\phi) - J_y(m_0) &= \mathbb{E} \int_0^T W(t) \overline{f(Y_t^\phi) - f(Y_t^0)} dt - \mathbb{E} \int_0^T e^{-\rho t} R_t(a_t^\phi) dt \\ &\quad - \chi e^{-\rho T} \mathbb{E} \{ \|\Pi Y_T^\phi\|_d^2 - \|\Pi Y_T^0\|_d^2 \}. \end{aligned} \quad (44)$$

Thus state-dependent consumption has to earn its utility-curvature cost through production or terminal dispersion. The identity is also an effective variance reduction: the common mean-state martingale cancels before simulation statistics are formed. Positive neural output is not itself evidence of positive economic improvement.

For each cell, calculate the scalar integrals

$$A_k = \int_{t_k}^{t_{k+1}} e^{-\rho t} dt, \quad B_k = \int_{t_k}^{t_{k+1}} W(t) dt,$$

$$C_k = \int_{t_k}^{t_{k+1}} e^{-\rho t} \{\log m_0(t) - w(t)m_0(t) - \zeta m_0(t)^2/2\} dt.$$

The consumption-deficit integral for a held action is exactly $C_k - A_k \overline{\log a_k} + B_k \bar{a}_k + \zeta A_k \bar{a}_k^2/2$. Consequently, consumption need not be approximated by a left-point utility sum. Let $X_{\phi,h}$ be the paired statistic obtained from (44) by using the internal Euler states in the production and terminal terms and these integrated consumption terms. The reference internal state uses the integrated common action $\int_{t_k}^{t_{k+1}} m_0(t) dt$. All scalar integral enclosures and their midpoint errors are saved.

8.3. A network-independent diffusion-transfer bound

Here the word independent concerns the transfer constants, not the economic statistic. Let

$$q_i = \sigma_I^2 \|B_i\|_2^2 + \sigma_C^2 \left(\sum_j B_{ij} \right)^2,$$

$$H = \kappa \left(d^{-1} \sum_i q_i \right)^{1/2}, \quad Q = \frac{\kappa}{2} \left(d^{-1} \sum_i q_i^2 \right)^{1/2}.$$

The letter Q here denotes a generator bound, not the dispersion bound Q_d in (34). For a policy in a radius- ϵ interval at sampling dates, put

$$M_\epsilon = \max_{m \in [m_0(0) - \epsilon, m_0(T) + \epsilon]} |c - m| + \kappa, \quad K_\epsilon = \beta M_\epsilon + Q.$$

The terminal dispersion majorant is

$$S_\epsilon = s_0 + (\kappa + \epsilon)T + \sigma_I \sqrt{(1 - 1/d)T}.$$

The formula uses the monotonicity of this economy's m_0 . For more general schedules their extrema replace the endpoints. Define

$$u_\epsilon(t) = h e^{\beta t} \{K_\epsilon t/2 + H \sqrt{t/3}\}, \quad (45)$$

$$\begin{aligned}
E_\epsilon(h) &= \beta \sum_k B_k u_\epsilon(t_k) + K_\epsilon \sum_k \int_{t_k}^{t_{k+1}} W(t)(t - t_k) dt \\
&\quad + \chi e^{-\rho T} u_\epsilon(T) \{2S_\epsilon + u_\epsilon(T)\}.
\end{aligned} \tag{46}$$

PROPOSITION 8—Transfer of paired policy values: *For exact arithmetic and untruncated internal innovations, the preceding construction satisfies*

$$|\Delta_\phi - \mathbb{E}X_{\phi,h}| \leq E_\epsilon(h) + E_0(h).$$

With clipped internal innovations, forward rounding, and enclosed scalar integration, replace each u_ϵ by its stated enlarged bound and add the statistic's quadrature and arithmetic error. Denote the resulting saved upper bound by $E_{\phi,h}^{\text{all}}$. It is $O(h)$ apart from these explicit numerical terms and does not involve a network derivative bound. The economic process remains the original continuous diffusion.

The proof is in the supplement. Its key step is to retain the martingale cancellation in the accumulated local drift error. Bounding each local stochastic error separately by its absolute magnitude would yield only an $O(\sqrt{h})$ account and would obscure the small policy differences studied here. This proposition concerns the sampled implementation (42); it does not prove this rate for arbitrary state-observed neural feedback.

8.4. A post-training certificate

Paired statistics are not uniformly bounded, because the terminal dispersion difference contains Gaussian shocks. This is dealt with explicitly, rather than ignored in a bounded concentration inequality. For each declared initial state and radius, the supplement constructs constants C_ϕ, A_ϕ such that

$$\Pr\{|X_{\phi,h}| > C_\phi + A_\phi(\sqrt{d-1} + v)\} \leq e^{-v^2/2}, \quad v \geq 0.$$

They are obtained from synchronous state separation and the norm of the projected reference innovations. Neither an empirical output range nor an empirical maximum enters their construction. With $v_* = 8$, put $b_\phi = C_\phi + A_\phi(\sqrt{d-1} + v_*)$

and $\tau_\phi = A_\phi e^{-v_*^2/2}/v_*$. Then clipping $X_{\phi,h}$ to $[-b_\phi, b_\phi]$ changes its expectation by at most τ_ϕ .

For n fresh independent paths, let $\bar{X}_\phi^c$ and s_ϕ^2 be the clipped empirical mean and unbiased variance. Fix a family of at most F one-sided statements before inspecting this noise bank, and set $\ell = \log(2F/\alpha)$. Define

$$L_\phi = \bar{X}_\phi^c - \sqrt{2s_\phi^2 \ell / n} - \frac{14b_\phi \ell}{3(n-1)} - E_{\phi,h}^{\text{all}} - \tau_\phi. \quad (47)$$

In the implementation the empirical mean is enclosed downward and the variance and every subtracted quantity upward. The upper payoff endpoint uses the same terms with reversed signs. The factor F counts both tails when both are reported.

THEOREM 9—Policy-specific continuous-time certificate: *Condition on the training and validation data, and suppose the specified final simulation paths are independent draws from the declared innovation law. Under the arithmetic contract and transfer assumptions above, with probability at least $1 - \alpha$, simultaneously for all declared policies and initial states,*

$$J_y(a^\phi) - J_y(m_0) \geq L_\phi, \quad 0 \leq V^*(y) - J_y(a^\phi) \leq G_d - L_\phi. \quad (48)$$

The optimum V^ still ranges over all adapted feasible controls in the original continuous economy, not merely the networks, schedules, or sampled controllers used in the experiment. Common shocks may be reused across policies; independence across those policies is not required. The guarantee is pointwise in each declared initial state and conditional on training, rather than uniform on an uncountable state domain.*

PROOF: Apply the empirical Bernstein inequality of [Maurer and Pontil \(2009, Theorem 4\)](#) to $(X_{\phi,h}^c + b_\phi)/(2b_\phi)$ and use a union bound over the declared one-sided statements. Rescaling gives the first two error terms in (47). The clipping expectation bound and Proposition 8 then give the first inequality in (48). Subtract it from the deterministic anchor bound (37). Feasibility yields nonnegative regret. An observed negative upper regret endpoint would indicate failure of the confidence event or of an implementation assumption, not a negative optimality gap. *Q.E.D.*

Unlike $G_d + D_{d,\epsilon'}$, the new endpoint depends on the deployed policy through its paired outcomes and sample variance. It can distinguish seeds and unsuccessful policies in the same architectural tube. When $L_\phi > 0$, the post-training guarantee is tighter than the analytical schedule's guarantee. This is a statement about the verified policy, not a deterministic promise that every training run improves welfare. The concentration inequality, the identity that transfers a lower improvement bound into a regret bound, and conservative policy improvement are established ideas (Kakade and Langford, 2002, Pirota et al., 2013, Scherrer et al., 2015). The contribution here is their operational connection to the continuous capital economy through (44)–(46), without a state-covering certificate or a global bound on the trained network's derivatives.

9. BELLMAN EVALUATION AND DIRECT METHOD COMPARISONS

The NBO map has two separate empirical obligations. Its evaluation block must produce a useful continuation-value derivative, and the resulting policy must justify the resources used to construct it. Verification of a policy does not by itself answer the second question. We therefore connect evaluation error to Hamiltonian error and compare candidate-generation methods by a direct paired payoff statistic. The economic problem and the feasible competitor class remain unchanged.

9.1. What the critic contributes

Use the normalized inner product $\langle x, z \rangle_d = d^{-1} x^\top z$ and norm $\|x\|_d^2 = \langle x, x \rangle_d$. At a state of the capital economy, set $q = dD_y V^\pi$. The part of the Hamiltonian depending on consumption is

$$H_q(m) = d^{-1} \sum_i \log m_i - \frac{\zeta}{2} \bar{m}^2 - \langle q, m \rangle_d.$$

On the original action box, and on any convex tube contained in it, H_q is μ -strongly concave for $\mu = \bar{m}_{\text{box}}^{-2}$. Here $\bar{m}_{\text{box}}$ denotes the primitive upper consumption-share bound, not the cross-sectional mean. Let m_q maximize H_q , let $\hat{q}$ be a critic costate, and let $\tilde{m}$ be within δ of the maximum of $H_{\hat{q}}$ over the same action set.

PROPOSITION 10—Evaluation error and feasible improvement: *Under the preceding conditions,*

$$H_q(m_q) - H_q(\tilde{m}) \leq \left(\sqrt{\delta} + \frac{\|q - \hat{q}\|_d}{\sqrt{2\mu}} \right)^2. \quad (49)$$

For a random rollout costate Z and a square-integrable predictor $g(X)$,

$$\mathbb{E}\|Z - g(X)\|_d^2 = \mathbb{E}\|Z - \mathbb{E}[Z | X]\|_d^2 + \mathbb{E}\|\mathbb{E}[Z | X] - g(X)\|_d^2. \quad (50)$$

These statements do not identify the costate of an Euler rollout with the continuous-time costate; a discretization or evaluation bias must be added when that identification is required.

Equation (49) identifies a reason to regress costates rather than only value levels: Hamiltonian loss is quadratic in a suitably measured costate error when the actor solves its improvement problem. Equation (50) identifies both the variance reduction available from conditional prediction and its regression bias. Neither identity promises that a trained critic reduces mean-square error relative to raw rollout derivatives. The new experiment therefore includes raw-costate improvement without a learned critic, value-only and costate-only critics, changes in block-update counts, and direct policy optimization. These are substantive alternatives, not alternative labels for NBO. The strong-concavity and conditional-projection arguments are established mathematical principles; their role here is to specify a testable evaluation–improvement mechanism, not to claim a new principle of policy iteration.

9.2. From evaluation error to economic improvement

Proposition 10 controls a decision at a given state. The distribution of states at which that decision is used matters for the economic conclusion. We make this link explicit. In this subsection, π is a fixed Markov policy with a sufficiently regular policy value V^π to justify Itô’s formula and the expectations below. Specifically, $V^\pi \in C^{1,2}([0, T) \times \mathbb{R}^d)$ is continuous at the terminal date with $V^\pi(T, \cdot) = g$. Along each candidate considered, the drift and terminal terms in the discounted Itô identity

are integrable and its stochastic integral is a zero-mean martingale. The condition

$$\mathbb{E} \int_0^T e^{-2\rho t} \|\Sigma^\top D_y V^\pi(t, Y_t^a)\|_2^2 dt < \infty$$

is sufficient for the latter. The parameters of π are frozen. Let a be any admissible candidate, including a history-dependent candidate, and let Y^a be its physical state. Set $q^\pi = dD_y V^\pi$ and define

$$g^\pi(t, y) = \max_{m \in \mathcal{A}} H_{q^\pi(t, y)}(m) - H_{q^\pi(t, y)}(\pi(t, y)) \geq 0,$$

where $\mathcal{A}$ is the same convex action set for both maximizations and both policies.

One may always take the primitive action box. For a costate predictor $\hat{q}$, let δ_t bound the $H_{\hat{q}(t, Y_t^a)}$ maximization gap of a_t over that set. Write $\bar{q}_h(t, y)$ for the conditional mean of the frozen-policy rollout costate at mesh h . Define, under the candidate's state law,

$$\mathcal{G}_a = \mathbb{E} \int_0^T e^{-\rho t} g^\pi(t, Y_t^a) dt,$$

$$\mathcal{D}_a = \mathbb{E} \int_0^T e^{-\rho t} \delta_t dt,$$

$$\mathcal{E}_{a,h}^2 = \mathbb{E} \int_0^T e^{-\rho t} \|\hat{q}(t, Y_t^a) - \bar{q}_h(t, Y_t^a)\|_d^2 dt,$$

$$\mathcal{B}_{a,h}^2 = \mathbb{E} \int_0^T e^{-\rho t} \|\bar{q}_h(t, Y_t^a) - q^\pi(t, Y_t^a)\|_d^2 dt.$$

The quantity $\mathcal{E}_{a,h}$ concerns regression accuracy; $\mathcal{B}_{a,h}$ retains the rollout discretization bias.

PROPOSITION 11—Evaluation, action quality, and payoff: *Suppose the displayed quantities are finite and the consumption Hamiltonian is μ -strongly concave on $\mathcal{A}$. Then*

$$J_y(a) - J_y(\pi) \geq \mathcal{G}_a - \left\{ \sqrt{\mathcal{D}_a} + \frac{\mathcal{E}_{a,h} + \mathcal{B}_{a,h}}{\sqrt{2\mu}} \right\}^2. \quad (51)$$

In particular, an upper bound on the action-loss term and a lower bound on $\mathcal{G}_a$ establish a sufficient condition for policy improvement. The proposition applies to any costate estimator;

a method comparison still requires the independently evaluated payoffs and their associated work.

PROOF: The fixed-policy equation for V^π , evaluated along Y^a , gives the performance-difference identity

$$J_y(a) - J_y(\pi) = \mathbb{E} \int_0^T e^{-\rho t} \{H_{q^\pi(t, Y_t^a)}(a_t) - H_{q^\pi(t, Y_t^a)}(\pi(t, Y_t^a))\} dt. \quad (52)$$

The diffusion and the state-dependent part of felicity are common to the two actions, so their differences cancel. The terminal terms agree. Localization followed by the assumed integrability justifies the martingale expectation. Subtract the true greedy Hamiltonian inside the integrand and use Proposition 10. Minkowski's inequality for the measure $e^{-\rho t} dt d\Pr$, first for $\sqrt{\delta_t}$ and the costate norm, and then for the regression and bias components of the costate, gives

$$\mathbb{E} \int_0^T e^{-\rho t} \left(\sqrt{\delta_t} + \frac{\|q^\pi - \hat{q}\|_d}{\sqrt{2\mu}} \right)^2 dt \leq \left\{ \sqrt{\mathcal{D}_a} + \frac{\mathcal{E}_{a,h} + \mathcal{B}_{a,h}}{\sqrt{2\mu}} \right\}^2.$$

Combining these statements proves the result. Q.E.D.

Independent costate diagnostics. Condition on a fitted policy, its critic, and a held-out state X . Let $\bar{Z}_1, \bar{Z}_2$ be means from independent rollout banks with the same conditional mean $\bar{q}_h(X)$. Conditional independence and zero conditional mean of the two errors imply

$$\mathbb{E}[\langle \hat{q}(X) - \bar{Z}_1, \hat{q}(X) - \bar{Z}_2 \rangle_d \mid X] = \|\hat{q}(X) - \bar{q}_h(X)\|_d^2.$$

With r independent replicates per bank and conditional single-rollout variance $v_h(X) = \mathbb{E}[\|Z - \bar{q}_h(X)\|_d^2 \mid X]$,

$$\frac{1}{2} \mathbb{E}[\|\bar{Z}_1 - \bar{Z}_2\|_d^2 \mid X] = \frac{v_h(X)}{r}.$$

The first sample average is not constrained to be positive. Both identities refer to the actual held-out state distribution. They estimate the quantities in (51) only when that distribution is the candidate's discounted state occupation law, or when a

justified change-of-measure account is supplied. Nested-grid differences diagnose time-step sensitivity; without a remainder bound they do not bound $\mathcal{B}_{a,h}$.

A continuous-time costate error account. There is also a direct sufficient condition for controlling the true costate, independent of a rollout bias estimate. For a smooth candidate value v , define its fixed-policy residual and terminal error by

$$r^\pi[v] = -v_t - \mathcal{L}^\pi v - u(\cdot, \pi) + \rho v, \quad b_v = v(T, \cdot) - g.$$

Suppose $b^\pi = c1 + f - \pi$ is continuously differentiable with $\|D_y b^\pi\|_2 \leq L_b$, and suppose differentiating the following Feynman–Kac expectations is justified. If $\|dD_y b_v\|_d \leq C_T$ and $\|dD_y r^\pi[v](s, \cdot)\|_d \leq C_r(s)$ everywhere, then

$$\|dD_y v(t, y) - q^\pi(t, y)\|_d \leq e^{(L_b - \rho)(T-t)} C_T + \int_t^T e^{(L_b - \rho)(s-t)} C_r(s) ds. \quad (53)$$

To see this, write $v - V^\pi$ as the expected discounted terminal error plus the expected discounted residual integral. The state Jacobian $J_{t,s}$ satisfies $\dot{J}_{t,s} = D_y b^\pi(s, Y_s) J_{t,s}$, $J_{t,t} = I$, and hence $\|J_{t,s}\|_2 \leq e^{L_b(s-t)}$. Differentiate the expectation and apply that bound. A hard terminal condition gives $C_T = 0$. Equation (53) is a sufficient mathematical account for a smooth continuous policy. A sampled residual, a rollout grid difference, or the derivatives of a different deployed implementation do not supply its global hypotheses.

9.3. Direct paired economic comparisons

Let a and b be two fitted sampled controllers, fixed before the final simulation bank. For each independent final path, compute their schedule-relative statistics $X_{a,h}$ and $X_{b,h}$ with exactly the same initial state and Brownian increments, and form

$$D_{ab,h} = X_{a,h} - X_{b,h}.$$

The common schedule’s numerical trajectory cancels. The implementation checks equality of its terminal arrays and the initial-state and innovation hashes. The deterministic transfer allowance is

$$E_{ab,h} = E_{a,h}^{\text{policy}} + E_{b,h}^{\text{policy}} + E_{a,h}^{\text{stat}} + E_{b,h}^{\text{stat}},$$

with outward arithmetic and a separate subtraction-roundoff cushion. In particular, it does not add the schedule transfer error twice. If b_a, b_b are the two clipping thresholds and τ_a, τ_b their expectation-tail allowances, use $b_{ab} = b_a + b_b$ and $\tau_{ab} = \tau_a + \tau_b$.

THEOREM 12—Simultaneous method and population comparisons: *Condition on all fitting and validation data. Suppose the final paths follow the declared independent sampling law and the conditional arithmetic and diffusion-transfer contract of Theorem 9. Apply the empirical Bernstein construction in (47) directly to $D_{ab,h}$, with threshold b_{ab} and allowances $E_{ab,h}, \tau_{ab}$. The resulting endpoints L_{ab}, U_{ab} satisfy*

$$L_{ab} \leq J(a) - J(b) \leq U_{ab}$$

simultaneously with the other declared one-sided statements at their joint confidence level. A difference of two schedule-relative lower endpoints is not used.

The same construction applies to $J_{\mathcal{P}}(a) = \sum_j p_j J_{y_j}(a)$ for a prespecified finite initial-state population $\mathcal{P}$. Draw its profile independently on every final path, average the conditional deterministic transfer bounds with the exact population weights, and use a common clipping threshold and tail bound valid for all profiles. The resulting guarantee is integrated over $\mathcal{P}$; it is not uniform over a continuum of initial states or over unobserved training seeds.

The comparison does not require NBO to dominate a direct method. It makes that empirical question answerable by the quantity actually of interest. A lower endpoint below zero is inconclusive about superiority unless the upper endpoint is also below zero. Certified improvement over the schedule and certified improvement over another trained policy remain different statements. Likewise, a smaller policy-regret upper endpoint need not equal the unknown actual regret.

10. OBSERVATION AND THE ECONOMIC IMPLEMENTATION

The capital state and the innovation-driven numerical controller are distinct. We now specify an observation model under which the controller uses the history of observed capital rather than latent economic shocks. The model assumes continuous noiseless observation of every capital coordinate, known diffusion and drift coefficients, knowledge of the controller's own actions, and independent private

randomization. It does not assume that a finite set of noisy observations reveals Brownian increments.

Write $\Sigma = [\sigma_I I_d, \sigma_C \mathbf{1}]$, $Q = \Sigma \Sigma^\top$, and let n be a unit vector spanning the one-dimensional null space of Σ . Since $\sigma_I > 0$, Q is invertible. The observed state path and known actions determine

$$M_t = Y_t - Y_0 - \int_0^t \{c\mathbf{1} + f(Y_s) - a_s\} ds.$$

Let B^0 be a scalar Brownian motion independent of the economic shocks and initial state, used only for private randomization.

PROPOSITION 13—Implementation from observed state history: *Define*

$$\tilde{W}_t = \Sigma^\top Q^{-1} M_t + n B_t^0. \quad (54)$$

Then $\tilde{W}$ is a $(d + 1)$ -dimensional Brownian motion in the enlarged observation filtration and $\Sigma \tilde{W}_t = M_t$. Feeding its cell increments into the prescribed numerical controller, including the componentwise clipping rule, yields the same closed-loop law and payoff as the innovation-driven controller. Thus its existing transfer and policy-specific performance statements apply under this observation model.

Private randomization in (54) is essential for this exact equivalence: the right-inverse term alone has rank d , whereas the implemented clipping acts separately on $d + 1$ drivers. The construction does not recover the original unobservable null component path by path. It replaces that component by an independent one with the correct law. Nor is the resulting controller a deterministic function of the state at a single decision date; its memory and independent randomization are part of its definition. The integrated drift in the proposition is an observation-model primitive. A sensor grid, drift quadrature or measurement error would need a further transfer allowance and is not silently treated as exact.

10.1. Information and the comparison class

The comparison class is defined on a canonical product space. Let $W = (W^1, \dots, W^{d+1})$ be the economic Brownian motion and let R be an independent

random element on a standard Borel space. The element R may contain a private Brownian path and a sequence of independent sensor errors. Write $\mathbb{F}^W$ for the usual augmentation of the economic Brownian filtration and $\mathbb{F}^{W,R}$ for the corresponding product filtration; the latter reveals only the portion of R specified by its observation clock. An admissible control is a progressively measurable, nonanticipative functional of the available paths, taking values in the primitive action box. Bounded controls and the globally Lipschitz drift give a unique strong state process and an integrable payoff. These conventions apply to the comparison class in Theorems 7 and 9.

PROPOSITION 14—Independent randomization and the optimum: *Let $V_W^*(y)$ and $V_{W,R}^*(y)$ be the expected-payoff suprema over the preceding economic and product filtrations. Then*

$$V_{W,R}^*(y) = V_W^*(y).$$

The anchor envelope G_d and the policy-specific regret conclusions therefore compare the observed-history and finite-observation policies with the same economic optimum. The equality concerns the supremum; it neither requires an optimal policy nor asserts that a single fixed randomizer realization attains that supremum.

PROOF: The inequality $V_{W,R}^* \geq V_W^*$ follows by ignoring R . For the reverse inequality, represent an admissible randomized control by its nonanticipative measurable functional $a(t, W_{[0,t]}, R_{[0,t]})$ before completion of the filtrations. Fixing a full realization r of R leaves a progressively measurable functional of W . For almost every r , Fubini's theorem preserves feasibility and the state equation; strong uniqueness identifies the resulting state with the controlled state for that section. Independence leaves the law of W equal to Wiener measure after conditioning on $R = r$. Thus the conditional payoff is at most $V_W^*(y)$ for almost every r . Integrating over r proves the reverse inequality. Null-set modifications in the augmented filtration do not change the state law or payoff. Alternatively, the anchor proof may be repeated directly on the product filtration: its synchronous comparison and martingale identities use only adaptedness, bounded actions, and the Brownian property, all of which are preserved. Q.E.D.

Exact continuous observation. For precision in Proposition 13, work first in the product filtration generated by W and an independent scalar Brownian motion B^0 . Let $\mathbb{G}$ be the usual augmentation of $\sigma(Y_s, B_s^0 : 0 \leq s \leq t)$, with the initial profile included. The stored numerical state and held actions are functions of this history. Consequently

$$M_t = Y_t - Y_0 - \int_0^t \{c1 + f(Y_s) - a_s\} ds$$

is $\mathbb{G}$ -adapted. Since it equals ΣW_t , it is a $\mathbb{G}$ -martingale: conditional expectation first in the product filtration and then in its subfiltration $\mathbb{G}$ proves this claim. With $A = \Sigma^\top (\Sigma \Sigma^\top)^{-1}$ and a unit null vector n , the process $AM + nB^0$ is a continuous $\mathbb{G}$ -local martingale starting at zero. Its bracket is $I_{d+1}t$, because $A\Sigma$ and $nn^\top$ are complementary orthogonal projections and $[M, B^0] = 0$. The multivariate Lévy characterization gives a Brownian motion in the stated observation filtration. The identity $\Sigma(AM + nB^0) = M$ holds pathwise.

This argument also establishes causality. At a decision date, the already observed drift integral determines the preceding innovation increment and hence the next internal state. The next action is then fixed before the next Brownian increment. The held-action state equation is uniquely solvable on that cell. Induction identifies the joint law of physical state, internal state, actions, and recovered innovations with the prescribed innovation controller. The independent null component supplies the correct law when clipping is applied separately to $d + 1$ coordinates; it does not recover the original latent null component. The exact observation result is separate from the finite measurement implementation below.

10.2. Finite observations with a continuous-economy transfer

At dates $t_k = kh$, a finite-observation controller receives $O_k = \bar{Y}_{t_k} + \eta_k$ and holds $\bar{a}_k = \hat{\pi}_k(O_k)$ until the next date. The sensor error is measurable at t_k in the product filtration just specified and satisfies $\|\eta_k\|_{2,d} \leq \nu_{\text{obs}}$, where $\|X\|_{2,d} = (\mathbb{E}\|X\|_d^2)^{1/2}$. Both the sensor state $\bar{Y}$ and the comparison parent's physical state Y obey the original continuous diffusion. No Brownian coordinate or continuous drift integral is supplied to the sensor controller.

The protected parent has internal state $Z_0 = y$ and selects $a_k = \hat{\pi}_k(Z_k)$. Its update is the specified clipped-innovation Euler update, with an implementation error of normalized norm at most ξ_h per cell. This includes innovation conversion and recurrence arithmetic. Both action maps use the same inward guard and lie within ϵ of the common schedule $m_0(t_k)$ coordinatewise. For a real reference map π_k , suppose

$$\|\pi_k(x) - \pi_k(z)\|_d \leq L\|x - z\|_d, \quad \sup_x \|\hat{\pi}_k(x) - \pi_k(x)\|_d \leq \delta_\pi.$$

Global matrix-norm products and forward error propagation through the protected activation supply L and δ_π for the saved weights. Measurement error is measured before the protected map; input representation within that map is included in δ_π .

Retain the production coefficient κ of the economic model and let $\beta \geq \kappa\|B\|_2$. Put $\bar{\sigma} = (\sigma_I^2 + \sigma_C^2)^{1/2}$ and let M bound $\|c1 + f(y) - m\|_d$ for all feasible held actions. Define the generator constants H and $K = \beta M + Q_f$ as in the diffusion-transfer argument of Theorem 9, where $H^2 \geq d^{-1} \sup_y \|Df(y)\Sigma\|_F^2$, $q_i = (B\Sigma\Sigma^\top B^\top)_{ii}$, and $Q_f = (\kappa/2)(d^{-1} \sum_i q_i^2)^{1/2}$ is the generator bound, not a diffusion covariance. Let $z > 0$ be the clipping threshold and let $v_{\text{clip}} \geq \mathbb{E}(G - \text{clip}(G, -z, z))^2$ for a standard normal G . The inherited bound $2\varphi(z)(z + z^{-1})$ is valid. Set

$$b_k = h\{Kt_k/2 + H\sqrt{t_k/3}\} + \bar{\sigma}\sqrt{t_k v_{\text{clip}}} + k\xi_h, \quad b_0 = 0, \quad (55)$$

$$e_0 = 0, \quad a_k^\Delta = \min\{2\epsilon, L(e_k + v_{\text{obs}}) + 2\delta_\pi\},$$

$$e_{k+1} = (1 + \beta h)e_k + ha_k^\Delta + (b_{k+1} - b_k), \quad 0 \leq k < N. \quad (56)$$

The forcing is nondecreasing. Each term retains the accumulated martingale cancellation; in particular, the clipping term is a finite-horizon second-moment account rather than a sum of cellwise absolute clipping errors.

Define the nonnegative function

$$D(t) = \int_0^t e^{\beta(t-s)} a_{\lfloor s/h \rfloor}^\Delta ds,$$

$$Q_T = s_0 + (\kappa + \epsilon)T + \sigma_I \sqrt{(1 - 1/d)T}.$$

Here $s_0 \geq \|\Pi Y_0\|_{2,d}$, with Π the projection away from the common-state direction.

For $t \in [t_k, t_{k+1})$, choose $m_{-,k} > 0$ below every action in the common tube and put

$$r_k(t) = \epsilon + |m_0(t) - m_0(t_k)|, \quad C_k(t) = r_k(t) \{[m_0(t)m_{-,k}]^{-1} + \zeta\}.$$

With $W(t) = e^{-\rho t} w(t)$ as before, define

$$\begin{aligned} B_{h,\nu_{\text{obs}}}^{\text{sens}} &= \beta \int_0^T W(t) D(t) dt + \sum_{k=0}^{N-1} a_k^\Delta \int_{t_k}^{t_{k+1}} e^{-\rho t} C_k(t) dt \\ &\quad + 2\chi e^{-\rho T} Q_T D(T). \end{aligned} \tag{57}$$

PROPOSITION 15—Finite-observation payoff transfer: *Under the stated assumptions,*

$$\|\bar{Y}_{t_k} - Z_k\|_{2,d} \leq e_k,$$

$$\|\bar{a}_k - a_k\|_{2,d} \leq a_k^\Delta,$$

$$\|\bar{Y}_t - Y_t\|_{2,d} \leq D(t),$$

and

$$|J_y(\bar{a}) - J_y(a)| \leq B_{h,\nu_{\text{obs}}}^{\text{sens}}.$$

Consequently, a valid improvement interval $[L_a, U_a]$ for the protected parent at the same time grid transfers to $[L_a - B_{h,\nu_{\text{obs}}}^{\text{sens}}, U_a + B_{h,\nu_{\text{obs}}}^{\text{sens}}]$ for the finite-observation controller. This deterministic operation uses no additional statistical confidence allocation. It requires a certificate for that exact protected parent and grid.

The allowance concerns the original physical diffusion and uses only the stated sensor moment bound. A shared-noise fine-Euler comparison checks the finite-observation implementation and reports its observed payoff difference. Such a diagnostic does not replace the allowance or certify the Euler bias. For fixed weights and horizon, uniformly bounded L, β, ϵ and generator constants, and $m_{-,k} \geq \underline{m} > 0$, the forcing vanishes if $h \rightarrow 0$, $v_{\text{clip}}(h) \rightarrow 0$, $\zeta_h/h \rightarrow 0$, $\nu_{\text{obs}} \rightarrow 0$, and $\delta_\pi \rightarrow 0$. Discrete Gronwall then makes the transfer allowance vanish. Fixed clipping and fixed numerical or measurement error leave the corresponding terms in the finite-grid bound.

10.3. *When is state-dependent management worth its fee?*

To give the policy differences a decision interpretation, consider a proportional resource cost $\tau \in [0, 1)$ for administering a state-dependent consumption rule. Gross withdrawals from capital remain $m_i e^{Y_i}$, and the service absorbs $\tau m_i e^{Y_i}$. Household consumption is $(1 - \tau)m_i e^{Y_i}$. Production, gross depletion, the gross-withdrawal adjustment cost, and terminal capital are unchanged. This is a self-financed resource experiment rather than an externally financed utility supplement. It changes the specified fee parameter, not the model used to train or benchmark the gross policy.

Let $A_T = \int_0^T e^{-\rho t} dt$. Log felicity gives the exact identity

$$J_{\mathcal{P}}^{\tau}(a) - J_{\mathcal{P}}(m_0) = J_{\mathcal{P}}(a) - J_{\mathcal{P}}(m_0) + A_T \log(1 - \tau). \quad (58)$$

Hence a positive certified improvement L_a supports every fee τ satisfying

$$\tau < 1 - \exp(-L_a / A_T).$$

A negative upper endpoint after the fee rules out an improvement for that candidate under the confidence event; an interval spanning zero is inconclusive. The fee grid is fixed before the final simulation. We do not equate wall-clock cost to a monetary fee or call these deliberately stylized balance sheets a calibrated household population. The experiment instead states precisely which adoption decisions are supported by the verified utility differences.

11. BUDGETED LEARNING, STATE COVERAGE, AND ECONOMIC DECISIONS

The new study separates the performance of a learned candidate from the incremental value of the Bellman evaluation block. The primitive capital economy, its action box, discounting, terminal payoff and dense coupling construction are unchanged. The fixed protocol uses dimensions 10, 20 and 50 and ten new initialization/minibatch streams. It is fixed after exploratory development and before the final fits and final simulation bank; it is not represented as a historical preregistration.

11.1. *Design and interpretation*

The primary population is a uniform distribution over nine balance-sheet profiles $y = \mu \mathbf{1} + s v_d$, with $\mu \in \{-0.5, 0, 0.5\}$, $s \in \{0, 0.25, 0.5\}$ and a fixed centered unit-root-mean-square linear profile v_d . The mean changes the aggregate level of initial capital, while s changes cross-sectional dispersion. This transparent design is distinct from both the single favorable origin and the deliberately severe state stresses. It is not fitted to data. Training and validation draw from this population and from subsequent-state perturbations. The final simulation independently redraws the initial profile and its future shocks. We also retain a separate origin comparison to make the change from the earlier study visible.

Each method receives the same announced 20-second fitting budget on the same worker for a given seed. The budget covers initialization, optimization, current-policy rollouts, validation, and checkpoint writes within the timed loop. The final selected-weight write and final evaluation lie outside this clock. The worker completes its current iteration; realized time and overshoot are reported. NBO performs policy evaluation with rollout value and costate targets, then feasible Hamiltonian improvement. Direct policy optimization differentiates the same actor's rollout payoff without a learned critic. The affine comparison uses a bounded affine actor. Validation, not final testing, selects the checkpoint. Architecture, action radius, training sampler and final economic problem are shared where the comparison calls for them. Differences in optimizer work are exposed by time, visits and update counts with their corresponding measurement boundaries.

Every primary final row uses 8,192 independent paths and 1,024 time cells. A family of 4,000 one-sided statements is allocated before final testing and covers both signs of every reported stochastic endpoint. Tables label the raw average as the mean paired statistic. The lower and upper endpoints include the empirical Bernstein margin, continuous-time transfer, tail clipping and arithmetic accounts. Across-seed ranges describe the ten recorded fitted objects; they are not confidence limits for every possible future training run.

TABLE II: Budgeted primary policy verification

d	Method / state	Mean	Lower	Regret ub.	Positive
10	NBO / 0	0.001982	0.001182	0.017565	10/10
10	DPO / 0	0.001906	0.001093	0.017654	10/10
10	LINEAR / 0	0.001868	0.001058	0.017689	10/10
20	NBO / 0	0.001586	0.000699	0.024826	10/10
20	DPO / 0	0.001495	0.000597	0.024929	10/10
20	LINEAR / 0	0.001456	0.000559	0.024967	10/10
50	NBO / 0	0.001755	0.000848	0.026692	10/10
50	DPO / 0	0.001657	0.000744	0.026796	10/10
50	LINEAR / 0	0.001523	0.000567	0.026972	10/10
10	NBO / P	0.001469	0.000628	0.020974	10/10
10	DPO / P	0.001382	0.000532	0.021070	10/10
10	LINEAR / P	0.001355	0.000510	0.021092	10/10
20	NBO / P	0.001346	0.000424	0.028470	10/10
20	DPO / P	0.001253	0.000326	0.028568	10/10
20	LINEAR / P	0.001227	0.000300	0.028594	10/10
50	NBO / P	0.001611	0.000664	0.030386	10/10
50	DPO / P	0.001520	0.000568	0.030482	10/10
50	LINEAR / P	0.001444	0.000471	0.030579	10/10

P denotes the fixed nine-profile population; 0 denotes the origin. Lower endpoints are rounded downward and regret bounds upward. Counts refer to fixed fitted policies, not unseen training seeds. All quantities except counts are in discounted utility units. The executed study contains 90 primary fitted policies, 244 policy-evaluation rows and 120 direct paired comparisons. For the prespecified population, 30 of 30 NBO policies have a positive simultaneous lower improvement endpoint. Of 30 population NBO-minus-direct-policy pairs, 0 have a positive lower endpoint and 0 have a negative upper endpoint; the remainder are inconclusive. These counts distinguish verified policy improvement from comparative method evidence. They do not turn

a completed optimization run, a positive sample mean or a loose global regret bound into proof of near optimality.

11.2. Does the Bellman block improve on direct optimization?

The relevant comparison is $J(a^{\text{NBO}}) - J(a^{\text{DPO}})$, not the difference between two separate lower bounds against the schedule. Table III reports direct common-path differences, their simultaneous continuous-time endpoints and the number of fitted pairs with a positive lower or a negative upper endpoint. Neither a positive raw mean nor a smaller sample standard deviation is substituted for this inference.

TABLE III: Direct NBO-minus-comparator endpoints

d	State	Comparator	Mean	Lower	Upper	+/-
10	0	DPO	0.000076	-0.001136	0.001294	0/0
10	0	LINEAR	0.000114	-0.001104	0.001329	0/0
20	0	DPO	0.000091	-0.001228	0.001414	0/0
20	0	LINEAR	0.000130	-0.001200	0.001455	0/0
50	0	DPO	0.000098	-0.001247	0.001441	0/0
50	0	LINEAR	0.000232	-0.001178	0.001618	0/0
10	P	DPO	0.000087	-0.001175	0.001351	0/0
10	P	LINEAR	0.000114	-0.001144	0.001372	0/0
20	P	DPO	0.000093	-0.001274	0.001462	0/0
20	P	LINEAR	0.000119	-0.001250	0.001491	0/0
50	P	DPO	0.000091	-0.001304	0.001484	0/0
50	P	LINEAR	0.000167	-0.001262	0.001583	0/0

Each row summarizes the recorded seed-matched pairs. The last column counts strictly positive lower endpoints and strictly negative upper endpoints. The other pairs are inconclusive. Bounds are computed from pathwise differences; the common schedule transfer cancels.

TABLE IV: Actual work at the announced fitting budget

d	Method	Fit sec.	Verify sec.	Actor ms.	Visits, mln.
10	NBO	20.04	15.62	0.339	1.987
10	DPO	20.04	15.62	0.342	2.896
10	LINEAR	20.02	8.88	0.154	3.918
20	NBO	20.04	23.83	0.376	1.835
20	DPO	20.04	23.47	0.378	2.613
20	LINEAR	20.03	16.51	0.188	3.432
50	NBO	20.05	47.56	0.454	1.511
50	DPO	20.05	47.77	0.453	2.048
50	LINEAR	20.04	41.49	0.272	2.492

Medians across the recorded primary fits. Verification refers to the population row. Actor timing is a batch of 256 proposals and excludes sensing and drift-integral acquisition. All fitting methods use the same announced 20-second budget; actual times include iteration completion. The saved verification clock starts after model loading and construction of deterministic interval constants. It is a timed simulation-and-reporting component, not a fully inclusive wall clock. Fitting clocks exclude the final selected-weight copy.

Training cost is only one component of work. The table reports verification separately because proving that a small difference is positive can be more expensive than constructing the candidate. Deployment timing concerns the actor proposal on a batch of 256 states; it excludes physical sensing, drift-integral observation and private-randomization infrastructure. It therefore must not be read as total implementation cost. The full record retains peak process memory, simulator-state visits, critic and actor updates, selected iteration, all checkpoint times and final evaluation costs.

The supplementary ablations remove the critic altogether by using raw rollout costates, remove either value-level or costate regression, change the number of block updates, and change network width. The radius frontier trains new actors at 0.05 and 0.15; it does not merely rescale a correction trained at 0.10. Checkpoint frontiers use the same full final path count as the primary study. Their selected checkpoints

are determined by the work clock, not by the final payoff. These experiments test the mechanism in Proposition 10 and expose cases where its potential variance reduction does not repay its approximation or optimization cost.

11.3. State stress and the price of implementation

Table V places the deliberately shifted and dispersed initial conditions in the main article. The protocol includes low and high aggregate capital, unit dispersion, and a joint low-mean/high-dispersion profile. These rows are not dropped when the raw difference is negative or the certified interval is inconclusive. The finite-population theorem does not extend a result on its nine profiles to these more severe states.

TABLE V: Severe initial-state stresses

d	Mean	Spread	NBO statistic	NBO lower	DPO lower
10	-2	0	0.000334	-0.000459	-0.000596
10	2	0	-0.000184	-0.000979	-0.001136
10	0	1	0.001048	0.000192	0.000114
10	-1	1.5	0.001638	0.000748	0.000701
20	-2	0	0.000316	-0.000565	-0.000602
20	2	0	-0.000196	-0.001078	-0.001300
20	0	1	0.001498	0.000547	0.000486
20	-1	1.5	0.001638	0.000653	0.000600
50	-2	0	0.000384	-0.000518	-0.000546
50	2	0	-0.000274	-0.001177	-0.001046
50	0	1	0.002010	0.001038	0.000953
50	-1	1.5	0.002256	0.001248	0.001206

Both methods are freshly fitted with the stated auxiliary seed under the population training design. These fixed-state stresses are outside the primary population in general. Negative lower endpoints are inconclusive, not proof of a negative true gain; full upper endpoints are saved.

TABLE VI: Self-financed management-fee support in the population

d	Method	5 bp	10 bp	20 bp	Floor, bp
10	NBO	10	0	0	6.40
10	DPO	10	0	0	5.43
10	LINEAR	10	0	0	5.20
20	NBO	0	0	0	4.32
20	DPO	0	0	0	3.32
20	LINEAR	0	0	0	3.06
50	NBO	10	0	0	6.77
50	DPO	10	0	0	5.79
50	LINEAR	4	0	0	4.81

The three central columns count fitted policies with a positive simultaneous lower improvement endpoint after the stated fee. One basis point is 0.01 percent of gross withdrawals. The ceiling is the smallest certified fee ceiling across the recorded seeds; zero does not prove that every positive fee is unprofitable.

The adoption account applies (58) at fees of zero, 0.05, 0.10 and 0.20 percent of gross withdrawals. A positive endpoint supports a particular fee-bearing policy in the specified population; a negative upper endpoint rejects its improvement relative to the fee-free schedule; a straddling interval does neither. This is a quantitative decision within the model, not an estimate of aggregate welfare or an empirical recommendation. Reporting an implementable resource threshold is more informative than describing every small utility difference as economically material.

11.4. Absolute policy accuracy in an independent nonlinear instance

The one-state member of the same model family admits a strong independent numerical reference without an exact homothetic or quadratic solution. We solve its monotone implicit HJB system on three nested grids and compare the largest grid with a wider state domain. The solver uses full-box feasible maximization and

banded linear solves; no neural critic enters its target values. Fixed-policy solves evaluate the learned Markov feedbacks on the same grid.

TABLE VII: Independent nonlinear scalar HJB policy-loss diagnostics

Space nodes	Time cells	L	NBO loss	DPO loss	Schedule loss
401	256	6	0.000003	0.000109	0.001543
801	512	6	0.000003	0.000114	0.001561
1601	1024	6	0.000003	0.000116	0.001570
1601	1024	8	0.000003	0.000115	0.001566

Losses are measured at initial log capital zero within the declared finite HJB approximation. The wider-domain row changes both boundary placement and mesh spacing. These are not certified multidimensional continuous-time optimality gaps.

The reference gives an absolute policy-loss diagnostic in a nonlinear instance and separates value accuracy from beating a feasible schedule. Its boundary approximation and finite-grid errors are exposed through domain and mesh sensitivity, not promoted to a rigorous continuous-domain certificate. The scalar grid feedback is also kept distinct from the innovation-history controller in the high-dimensional theorem. The latter’s global optimum comparison remains the explicit $G_d - L_a$ account. An independent scalar reference does not itself prove that a 50-dimensional neural policy is near optimal.

11.5. *Scope of the computational claim*

The evidence identifies several separate objects: the NBO evaluation and improvement map, an action-loss bound conditional on its evaluation accuracy, policy-specific economic verification, direct method comparisons, and the cost of a specified implementation decision. None of them makes universal superiority a premise. The capital transfer and observation propositions are model-specific. Recursive utility, endogenous preference adjustment, stopping, temporal selves and strategic deviations retain the models and verification accounts developed elsewhere in this

paper and its supplement; their hypotheses are not silently supplied by the capital proof.

12. WORK, COSTATE ACCURACY, AND IMPLEMENTED DECISIONS

The fixed-work experiment complements the announced-time study with a comparison whose simulator workload does not depend on the worker’s speed. It uses the same economic primitives and action radius, dimensions ten, twenty, and fifty, and two specified training streams. Each method runs eighty iterations. The exact twentieth and eightieth checkpoints are evaluated on a fresh bank of 8,192 paths and 1,024 cells. NBO, direct policy optimization, the affine actor, and raw-costate improvement use the same number of training states and rollout cells. The raw-costate method differentiates current-policy rollouts and improves an actor without fitting a critic. Its action class and rollout work agree with NBO’s; its evaluation work does not include critic regression.

12.1. Fixed work and attained economic precision

One iteration uses 128 states and 32 rollout cells. Validation occurs every twenty iterations with 128 paths and 64 cells. The twentieth checkpoint therefore uses 81,920 training-state visits and 8,192 validation-state visits; the eightieth uses 327,680 and 32,768, respectively. NBO and raw-costate improvement perform five actor updates per iteration, while the two direct methods perform one. NBO also performs five critic updates. Equal simulator visits and equal iteration counts consequently do not imply equal floating-point work. The recorded update counts and clocks expose that distinction.

TABLE VIII: Fixed-iteration and simulator-visit extension

d	k	Method	Mean	Min. lower	Positive	Fit sec.
10	20	NBO	0.001381	0.000539	2/2	1.58
10	20	DPO	0.000998	0.000157	2/2	0.59
10	20	Affine	0.000530	-0.000301	0/2	0.43
10	20	Raw	0.001435	0.000595	2/2	0.68

d	k	Method	Mean	Min. lower	Positive	Fit sec.
10	80	NBO	0.001443	0.000604	2/2	4.22
10	80	DPO	0.001346	0.000504	2/2	2.36
10	80	Affine	0.000858	0.000022	2/2	1.72
10	80	Raw	0.001456	0.000616	2/2	2.72
20	20	NBO	0.001283	0.000356	2/2	1.77
20	20	DPO	0.000929	0.000009	2/2	0.73
20	20	Affine	0.000665	-0.000257	0/2	0.54
20	20	Raw	0.001311	0.000390	2/2	0.83
20	80	NBO	0.001321	0.000400	2/2	4.92
20	80	DPO	0.001219	0.000295	2/2	2.93
20	80	Affine	0.000879	-0.000042	0/2	2.15
20	80	Raw	0.001334	0.000413	2/2	3.30
50	20	NBO	0.001532	0.000580	2/2	1.92
50	20	DPO	0.001190	0.000221	2/2	0.86
50	20	Affine	0.000874	-0.000081	0/2	0.71
50	20	Raw	0.001557	0.000610	2/2	0.97
50	80	NBO	0.001582	0.000634	2/2	5.50
50	80	DPO	0.001436	0.000491	2/2	3.43
50	80	Affine	0.001089	0.000141	2/2	2.87
50	80	Raw	0.001595	0.000650	2/2	3.88

Ubuntu 24.04; both fixed training seeds are retained. Each method uses 81,920 training and 8,192 validation state visits at 20 iterations, and 327,680 plus 32,768 at 80. NBO and Raw use five actor updates per iteration; DPO and Affine use one. NBO additionally uses five critic updates. Fit seconds are checkpoint-prefix clocks. Means are numerical paired statistics; lower endpoints include the stated continuous-time transfer and sampling account.

TABLE IX: Certified targets and accounted work

d	Method	$g = 0$	Sec.	$g = .0005$	Sec.	$g = .001$	Sec.
10	NBO	2/2	20.9–21.2	2/2	20.9–21.2	0/2	–
10	DPO	2/2	18.2–18.5	2/2	34.0–34.7	0/2	–

d	Method	$g = 0$	Sec. $g = .0005$	Sec. $g = .001$	Sec.
10	Affine	2/2 19.7–20.2	0/2	–	0/2 –
10	Raw	2/2 17.2–19.3	2/2 17.2–19.3	0/2	–
20	NBO	2/2 30.1–30.9	0/2	–	0/2 –
20	DPO	2/2 27.4–27.8	0/2	–	0/2 –
20	Affine	0/2 –	0/2	–	0/2 –
20	Raw	2/2 26.3–28.1	0/2	–	0/2 –
50	NBO	2/2 56.2–58.1	2/2 56.2–58.1	0/2	–
50	DPO	2/2 51.5–55.7	0/2	–	0/2 –
50	Affine	2/2 86.8–92.4	0/2	–	0/2 –
50	Raw	2/2 52.9–54.9	2/2 52.9–54.9	0/2	–

Each count is the number of the two fixed seeds whose 20- or 80-iteration checkpoint has a lower endpoint strictly above the target. Seconds give the range of minimum executed method-accounted work among attaining seeds, charging the full 80-iteration fit and every verification through the first qualifying checkpoint. A dash means no tested checkpoint attained the target. These are recorded computational components, not complete end-to-end wall clocks; Table VII also gives retrospective checkpoint-prefix accounts and every failure to attain a target.

The common targets are schedule-improvement lower endpoints of zero, 0.0005, and 0.001. For each fitted stream, we inspect the declared checkpoints in temporal order and include verification of every earlier unsuccessful checkpoint. Each full verification advances both a policy and the common schedule for 16,777,216 state transitions. Reaching the second checkpoint therefore incurs two such verification calculations in addition to the training and validation visits. The full supplementary table retains unreached targets and reports the available clocks without assigning an artificial zero cost to failure.

There are two distinct timing interpretations. The executed experiment trained all eighty iterations before either checkpoint was evaluated; its recorded cost includes that complete fit. A retrospective prefix frontier uses the saved clock at the tested checkpoint and asks what an otherwise identical sequential procedure would have

consumed. This frontier is informative about work allocation, but is not a measured early-stopping speedup. Moreover, the saved evaluation clock begins after weight loading and deterministic interval-account construction. Sums of those clocks measure recorded timed components. Exact simulator visits, update counts, full worker records, and the omitted clock boundaries are reported alongside them.

The second operating-system environment repeats dimension ten and the first fixed stream. It provides an execution-sensitivity check with the same source, nominal seeds, protocol, and work budgets. The initial-state hashes agree, but the saved innovation hashes differ across environments. Consequently, its endpoint differences are not treated as a common-path cross-environment estimator. Within each environment, every method comparison still uses exactly matching innovation and initial-state hashes. The supplement reports the weight, clock, and endpoint changes without treating the repeated environment as an additional optimizer-population observation.

12.2. *The observed evaluation-improvement mechanism*

The independent mechanism panels use sixty-four held-out states and sixteen nested rollout replicates, with thirty-two and 128 time cells. A critic predicts a costate for the frozen policy. Coarse rollout replicate zero supplies the raw-costate action. The common reference for the corrected comparisons is the mean of fine replicates one through fifteen, which is independent of that raw action’s replicate and of the fitted critic. All three saved actions are evaluated against this same reference, and its constrained greedy action is recomputed from the primitive Hamiltonian. The original diagnostics and arrays are preserved together with the corrected reference calculation.

TABLE X: Held-out costate errors on a common independent reference

d	Seed	Critic MSE	Raw MSE	Reference var.	Nested change	Nested var.
10	7919	6.946	6.560	0.491	0.048	0.000203
10	15401	8.036	6.542	0.490	0.048	0.000203

d	Seed	Critic MSE	Raw MSE	Reference var.	Nested change	Nested var.
20	7919	7.886	5.824	0.367	0.037	0.000125
20	15401	7.536	5.840	0.368	0.037	0.000125
50	7919	11.445	8.392	0.445	0.047	0.000177
50	15401	16.738	8.413	0.445	0.046	0.000179

All displayed entries are multiplied by 10^5 . Costates are scaled as $dD_y V$. Both critic and raw MSE use the same mean of 15 independent 128-step rollouts, excluding the replicate coupled to the raw 32-step sample. Reference variance is the estimated variance of that mean. Nested change is the mean squared difference between the 16-replicate coarse and fine means; nested variance uses paired replicate differences. These are held-out Euler diagnostics; subtracting the corresponding Monte Carlo variance gives the signed debiased diagnostic saved in the audit, not a continuous-time bias certificate.

TABLE XI: Independent-reference Hamiltonian gaps and the raw-policy contrast

d	Seed	Actor gap	Critic gap	Raw gap	Payoff mean	Lower	Upper
10	7919	3.245	1.161	1.027	-1.290	-126.505	123.925
10	15401	3.210	1.338	1.024	-1.280	-126.496	123.936
20	7919	3.318	1.387	0.980	-1.231	-137.511	135.049
20	15401	3.710	1.328	0.982	-1.294	-137.565	134.977
50	7919	4.306	1.877	1.253	-0.915	-139.585	137.756
50	15401	4.859	2.723	1.255	-1.565	-140.252	137.122

All displayed entries are multiplied by 10^5 . Gaps evaluate the original actor, critic-greedy action, and raw-greedy action against an independently solved tube optimum using the common 15-replicate fine-costate mean. The final three columns are separate certified NBO-minus-Raw policy comparisons at 80 iterations. The Raw policy is trained independently of the frozen NBO-policy diagnostic. These columns therefore display distinct links in the proposed mechanism, without a causal or occupancy-weighted identification claim.

In all six principal dimension–stream panels, the critic’s mean squared error exceeds the raw-costate error against the common independent reference. The ratio ranges from approximately 1.06 to 1.99. The critic-implied greedy action also has a larger reference Hamiltonian gap in every panel. These observations locate a restriction of the fitted evaluation block: its regression does not realize the variance-reduction opportunity in Proposition 10 for these states, budgets, and networks. They do not alter the independently verified policy payoffs. The direct NBO-minus-raw economic intervals in the accompanying table assess the resulting policies as a separate question.

The reference mean remains noisy and discretized. Its Monte Carlo variance and the change between nested rollout grids are reported separately. The held-out states follow the declared training-state design, rather than the improved policy’s discounted occupation law. Thus these panels test the evaluation mechanism without substituting their empirical mean squared errors into the welfare inequality in Proposition 11. That proposition supplies the additional bias, occupation, and action-optimization terms needed for such a substitution.

12.3. *Independent reference and observation diagnostics*

The scalar calculation is replayed with the original frozen policies and all four original grids. An independent stencil calculation records each fixed-policy equation residual, the Howard residual and iteration count, boundary values, and action-bound frequencies at every time step. The replay reproduces all original arrays exactly in the recorded validation environment. The supplementary tables give value and action changes under refinement and policy losses at five distinct initial states.

On the finest domain-six grid, the NBO loss at initial state zero is approximately 3.44×10^{-6} , compared with 1.16×10^{-4} for direct policy optimization. At initial state one, the corresponding losses are approximately 1.05×10^{-4} and 5.02×10^{-5} . The off-center comparison therefore differs from the center comparison. Widening the domain from six to eight also changes the spatial step, and is reported as joint domain-and-spacing sensitivity. The high-dimensional optimum comparison

continues to use its own explicit $G_d - L_a$ bound; scalar losses are not extrapolated to another dimension or controller.

TABLE XII

SCALAR REFERENCE: EQUATION RESIDUALS AND HOWARD CONVERGENCE

n_y	n_t	L	HJB residual	Linear residual	Iter./step	Replay error
401	256	6	1.73×10^{-12}	1.86×10^{-12}	3-3	0
801	512	6	3.50×10^{-12}	3.57×10^{-12}	3-3	0
1601	1024	6	8.30×10^{-12}	8.64×10^{-12}	3-3	0
1601	1024	8	1.19×10^{-11}	1.33×10^{-11}	3-3	0

Residuals are maxima over every interior state and every time step. The linear column also includes every Howard linear solve. All time steps satisfy the original 10^{-11} value-iterate stopping rule. Replay error is the largest absolute discrepancy across all original arrays; it is a floating-point reproducibility check, not an economic approximation error.

TABLE XIII

SCALAR FIXED-POLICY RESIDUALS AND ACTION-BOUND FREQUENCIES

Grid (n_y, n_t, L)	NBO	DPO	Affine	Anchor	Max. bound freq.
(401,256,6)	1.86×10^{-12}	1.81×10^{-12}	1.76×10^{-12}	1.77×10^{-12}	0.00%
(801,512,6)	3.43×10^{-12}	3.56×10^{-12}	3.50×10^{-12}	3.45×10^{-12}	0.00%
(1601,1024,6)	8.38×10^{-12}	8.30×10^{-12}	7.77×10^{-12}	8.49×10^{-12}	0.00%
(1601,1024,8)	1.31×10^{-11}	1.28×10^{-11}	1.33×10^{-11}	1.33×10^{-11}	0.00%

Method columns give maximum fixed-policy equation residuals. The last column is the largest bound frequency among the five policies, counting either endpoint of the unchanged economic action interval at all interior state-time nodes, with absolute tolerance 10^{-12} . These are unweighted node frequencies, not visitation probabilities. Every imposed Dirichlet boundary value is satisfied by construction; its error relative to the unbounded economy is not enclosed. Method-specific counts, denominators, action ranges, time-zero actions and residual traces are retained in the diagnostic files.

TABLE XIV: Finite-observation implementation and transfer

d	Cells	ν_{obs}	Euler difference	Transfer ub.
10	64	0	6.049e-09	0.001271
10	64	0.001	5.781e-09	0.002923
10	64	0.01	-2.205e-08	0.017790
10	128	0	7.056e-10	0.000635
10	128	0.001	8.911e-10	0.002266
10	128	0.01	-2.375e-08	0.016941
20	64	0	-2.751e-09	0.002751
20	64	0.001	-3.501e-09	0.006207
20	64	0.01	-2.421e-08	0.037305
20	128	0	-1.545e-09	0.001386
20	128	0.001	-1.123e-09	0.004824
20	128	0.01	-1.119e-08	0.035766
50	64	0	-1.520e-09	0.008369
50	64	0.001	-1.543e-09	0.020831
50	64	0.01	-6.657e-09	0.075404
50	128	0	-1.633e-09	0.004314
50	128	0.001	-1.828e-09	0.017004
50	128	0.01	-8.270e-09	0.073836

The fourth column is a 128-path shared-noise fine-Euler mean; it has no continuous-time confidence interpretation. The last column is the separate outward continuous-diffusion transfer allowance, displayed upward. Every decision cell has eight fine diagnostic cells. No 1,024-cell parent certificate is assigned to these coarser parents.

The finite-observation experiment implements the direct measured-state policy of Proposition 15. It fixes the original eightieth NBO checkpoint in each dimension and uses 128 paths, sixty-four or 128 decision cells, and eight fine Euler cells per decision. Independent sensor noise has normalized root-mean-square levels zero, 0.001, and 0.01. Both compared controllers use the same protected saved-weight map. Their shared-noise payoff differences are finite-Euler diagnostics. The separate

outward transfer calculation accounts for sensing, the internal recurrence, network arithmetic, and clipping in the original continuous diffusion. No diagnostic difference is substituted for a continuous-time endpoint, and no certificate from the 1,024-cell experiment is assigned to a different parent or grid.

These comparisons complete the evidence for the paper's economic distinctions. A policy may have a positive verified gain over a feasible schedule while its ranking against another trained policy remains statistically unresolved. A useful scalar approximation may have state-dependent relative accuracy. A critic may supply a coherent improvement direction while a cheaper costate estimator performs better on the declared diagnostic. The NBO error account states the quantities needed to assess each of these outcomes within the original economic problem.

13. CONCLUSION

Neural Bellman Operators provide an evaluation and improvement structure for continuous-time economic control and equilibrium problems. The critic approximates the continuation value associated with a specified policy; the actor makes feasible choices using that value; and independent verification measures the remaining economic error. Differential verification, monotone Bellman approximation, and complete-action enclosures make these roles explicit across the models considered here.

The capital application connects a fitted policy to the original diffusion through a policy-specific payoff comparison. The analytical schedule supplies an absolute benchmark, and direct paired inference measures the gain from a particular learned policy after accounting for implementation and sampling error. Costate accuracy provides a mathematical link between critic evaluation and Hamiltonian improvement. The computational comparisons assess that link against direct policy optimization and other feasible alternatives, while reporting the work required both to construct and to verify each candidate.

The economic program remains the one that motivates NBO. Endogenous preferences combine consumption, portfolio risk, and costly internal adjustment. Recursive utility changes the continuation-value equation. Temporal selves and competing firms require equilibrium-specific deviation conditions. Their applications retain

their own models, derivations, economic comparisons, and verification results. The resulting framework makes quantitative claims assessable in utility, compensation, and unilateral-profit units, with the information structure and approximation domain stated for each claim.

The theoretical and computational results identify where further precision is valuable: policy evaluation must improve the derivatives relevant for decisions, and the resulting gain must justify the total work and implementation cost. This connection between a numerical Bellman calculation and a verified economic decision is the organizing contribution of the paper.

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